



Recent Advances in Antibiotic Degradation via Sulfate Radical Mediated AOPs: Bond Cleavage Mechanism, Fragmentation Pathways, and Interfacial Chemistry

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Abstract Sulfate radical-based advanced oxidation processes (SR-AOPs) have emerged as promising and efficient technologies for the removal of antibiotics (ATBs) from aquatic environments due to their ability to generate highly reactive oxygen species and achieve effective contaminant degradation. The increasing persistence of ATBs in water bodies and their associated ecological and human health risks have stimulated extensive research on SR-AOP-based treatment strategies, including sulfate radical-driven oxidation, photocatalysis, electrochemical oxidation, sono-chemical processes, ozonation, and integrated hybrid systems. However, a comprehensive understanding of the molecular-level degradation mechanisms of ATBs, particularly the identification of reactive sites, bond cleavage pathways, and the roles of radical and non-radical reactive oxygen

species ($\text{SO}_4^{\bullet-}$ / $^{\bullet}\text{OH}$ / $\text{O}_2^{\bullet-}$ / $^1\text{O}_2$) remains insufficiently explored. This review provides a critical assessment of the sources, environmental occurrence, exposure pathways, and potential impacts of ATBs, along with existing regulatory frameworks for controlling antibiotic pollution. Furthermore, recent advances in SR-AOP strategies for ATB degradation are systematically discussed, with emphasis on reaction mechanisms, transformation products, structure reactivity relationships, and molecular fragmentation pathways. The review also highlights key challenges associated with catalyst development, oxidant utilization, energy consumption, and economic feasibility for large-scale applications. Finally, future research directions are proposed to promote sustainable catalyst design and accelerate the transition of SR-AOP technologies from laboratory investigations toward practical wastewater treatment applications. This review offers new mechanistic insights into ATB degradation and

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provides guidance for developing efficient and sustainable SR-AOP-based remediation strategies.

Keywords Antibiotics degradation · Reactive oxygen species · Persulfate activation · Sulfate radical · Electrophilic and nucleophilic attack · Transformation pathways

Abbreviation

SR-AOP	Sulfate radical-based advanced oxidation processes
ATBS	Antibiotics
ARGs	Antibiotic resistance genes
ROS	Reactive oxygen species
PMS	Peroxymonosulfate
PDS	Peroxydisulfate
H ₂ O ₂	Hydrogen peroxide
EF	Electro-Fenton
PEC	Photoelectrochemical process
US	Ultrasound
DFT	Density functional theory
QSAR	Quantitative structure–activity relationship
SMX	Sulfamethoxazole
CIP	Ciprofloxacin
OFLX	Ofloxacin
TC	Tetracycline
AMX	Amoxicillin

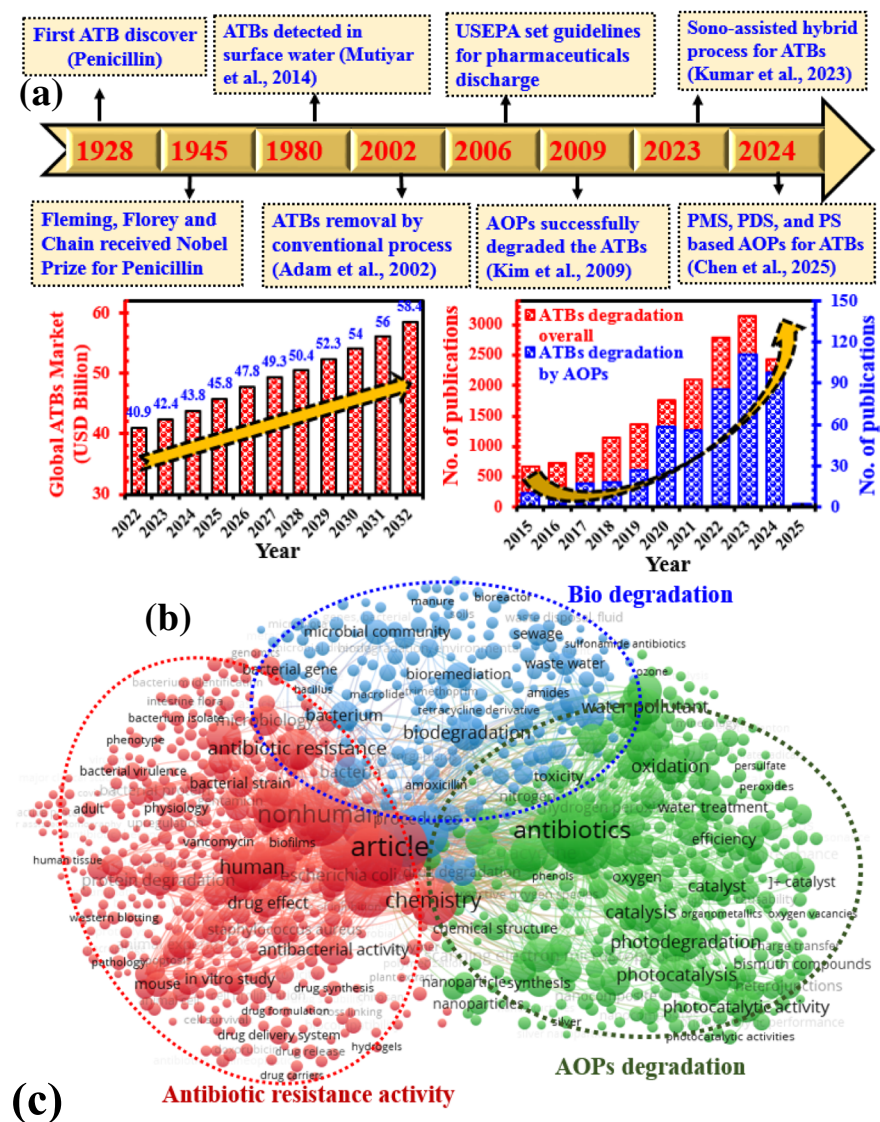
1 Introduction

The rapid increase in wastewater generated associated with pharmaceutical manufacturing in recent decades has garnered attention due to its detrimental effects on aquatic life and the hazards it presents to human health (Changotra, Guin, et al. 2019; P. Gao et al., 2021). As a result, drugs' environmental impact has grown to be a major concern. Because of their rapid production rates and extensive use, antibiotics have drawn special attention among the large range of pharmaceuticals (Hassani et al., 2023). Due to their extensive usage in agriculture, veterinary care, and medicine, antibiotics have become chronic environmental contaminants (Arvind Kumar et al., 2023a, 2023b). Certain types, such as fluoroquinolones and tetracyclines, are extremely resistant to breakdown, and even at low concentrations, they

contribute to antibiotic resistance and provide difficulties for traditional wastewater treatment (Yuan et al., 2023).

Figure 1a shows the research timeline on antibiotic detection, degradation, and worldwide manufacturing as well as papers over the last ten years on antibiotic degradation through AOPs that were found using Google Scholar (keyword search: "antibiotics degradation," August 2025). The literature trends that were taken from the Scopus database using the keywords "antibiotics degradation" and "advanced oxidation processes" are shown in Fig. 1b. Additionally, Fig. 1c offers a summary of how medications are categorized and how they affect human health. Antibiotics are commonly used to treat infectious diseases and infections, as well as to promote growth and reduce production cycles in aquaculture and livestock. However, because many microorganisms have developed several kinds of resistance, their efficacy has been progressively compromised (Nabilah Mohd Noor et al., 2023). Because drugs promote antibiotic-resistant bacteria (ARB) and antibiotic resistance genes (ARGs), their presence in aquatic settings is a serious worldwide problem (Qin et al., 2021). Antibiotic-resistant bacteria (ARB) and genes (ARG) proliferate when antibiotics are discharged into rivers and lakes, endangering ecosystems and human health. The effectiveness of medical therapies may be diminished due to the presence of these resistant bacteria which can be transferred from one person to another through food (Iqbal et al., 2024). The overuse of antibiotics, failure to complete prescriptions, incorrect diagnoses, and their use in agriculture for growth promotion are the key factors in producing environmentally resistant bacteria (Kanakaraju et al., 2018). According to World Health Organization (WHO), approximately 80% of the total antimicrobial consumption in animal agriculture is antibiotics (Changotra, Rajput, et al., 2019a, 2019b). The WHO stated that, antibiotic resistance is a complicated, multifaceted phenomenon with many contributing factors, and the adoption of isolated solutions to the problem has not been very effective (Krasucka et al., 2021). The WHO does not yet have targeted actions to deal with environmental routes of antibiotic resistance. As indicated in Fig. 1a, the production of antibiotics has increased exponentially, from approximately 288 million tons annually to a projected 33 billion tons in 2050 (Nabilah Mohd Noor et al., 2023).

Fig. 1 **a:** Milestone time-line of research related to ATBs, global ATBs production, last 10-year publications on ATBs degradation by AOPs (Google scholar, antibiotics degradation as keyword search made in august 2025), **b** Existing literature based on Scopus database searched by keyword “antibiotics degradation” and “advanced oxidation processes”, and **c** classification of pharmaceuticals and their impact on human



Pharmaceuticals/Class	Impacts on human health and environment
Antibiotics	▪ Increase oxidative stress, block the electron chain of photosynthesis, developed emerging resistance, affect to algae and aquatic plant
Anti-cancer	▪ Genotoxic, mutagenic, cytotoxic, and teratogenic in eukaryotic organism, genetic abnormality, and visceral organ damage
Endocrine disruptors	▪ Alzheimer diseases, cancer, obesity, and thyroid disturbance
Analgesics and NSAIDS	▪ Breeding pattern changes in <i>Oryzias latipes</i> , and inhibit the CYP2M in <i>Cyprinus carpio</i>
Anti-retroviral	▪ Nephrotoxicity, nausea, rashes, anemia, and hypersensitivity
Bitra-blockers	▪ Mortality in the fish and algae, embryonic development issues in crustacean, and immobilization in <i>Daphnid similis</i>

In order to halt the progression of AMR and protect human and environmental health, we must take action now to address this escalating problem (Krasucka et al., 2021). As a result of this, many countries are putting laws and regulations in place to restrict the use of antibiotics. To facilitate the development of effective therapies and globally protect the public's health, it is vital to build an understanding of the environmental routes of AMR (Sosa-Hernández et al., 2021). This article highlights the critical need for innovative strategies to mitigate antibiotics (ATBs) due to the dramatic increase in publications over the past ten years. This pattern can be seen in Fig. 1a. The rapid and sustainable treatment methods are crucial for degrading ATBs from wastewater given the complexity and gravity of the problem. Conventional techniques include adsorption, biodegradation, incineration, chlorination, and reverse osmosis just move pollutants between phases; they are less advantageous since they frequently result in secondary contamination, are impacted by pollutant toxicity, and have high operating costs.

The degradation of ATBs from wastewater employing different AOPS technologies, such as sulfate radical-based AOPs (SR-AOPs) (Arvind Kumar et al., 2023a, 2023b), photocatalysis (Du et al., 2023; Hassani et al., 2023; Xindan Zhang et al., 2024a, 2024b), Fenton oxidation (X. Sun et al., 2023), electrooxidation (Farissi et al., 2023), ozonation (Noman et al., 2024), and sono-chemical process (Karim & Shriwastav, 2021; B. Liu et al., 2021a, 2021b), is currently receiving more and more attention. Due to a number of benefits, sulfate radical-based AOPs (SR-AOPs) are becoming more popular among AOPs than hydroxyl radical-based systems ($\cdot\text{OH}$ -AOPs). Because sulfate radicals ($\text{SO}_4^{\cdot-}$) have a longer half-life (30–40 μs) and a greater redox potential (2.5–3.1 V) than $\cdot\text{OH}$, they can interact with contaminants more effectively and sustainably. Furthermore, SR-AOPs function well across a greater pH range, and the precursors used to produce $\text{SO}_4^{\cdot-}$ peroxymonosulfate (PMS) and peroxydisulfate (PDS) are more stable than those used for $\cdot\text{OH}$ (e.g., H_2O_2 , O_3). SR-AOPs are a reliable and scalable method for removing antibiotics from contaminated waters since these processes can either completely mineralize ATB molecules into CO_2 and H_2O or break them down into tiny intermediates. The study of ATB degradation by AOPs has benefited greatly from recent

review studies. For example, in a variety of AOP systems, (F. Li et al., 2024a, 2024b) investigated catalytic sites and the related processes of catalytic antagonistic genes. In another recent review, (Yanxin Li et al., 2024a, 2024b) focused primarily on material innovations including carbon-based materials, covalent organic frameworks (COFs), metal–organic frameworks (MOFs), and transition metals for persulfate activation and their roles in ATB degradation. Nevertheless, a thorough mechanistic understanding and their degradation mechanisms for ATBs specifically via SR-AOPs have not yet been presented in any review study. Important elements including molecular fragmentation, the identification of reactive attacking sites (electrophilic, nucleophilic, and radical attacking sites) on the ATBs molecule, and the detailed mechanism of ATBs molecule degradation by AOPs are still not well characterized. In addition to being crucial for expanding scientific knowledge, filling in these gaps directly supports the Sustainable Development Goal (SDG) 2030 of the United Nations, especially with regard to preserving water quality, lowering environmental pollution, and safeguarding human health from pharmaceutical contaminants. To address this critical gap, the present review aims to: (a) briefly review the sources, occurrence, human exposure routes, adverse effects of ATBs in the environment, and legislations aimed at mitigation of ATBs pollutants; (b) examines cutting-edge frontline strategies in several SR-AOPs strategies for ATBs elimination; with emphasis on by-product generation, degradation pathways, and in-depth mechanistic discussions concerning the ATBs structural breakdown; and (c) address the existing challenges and outline future research directions for eliminating ATBs through AOPs.

1.1 Literature Search Strategy

To accomplish the objectives outlined above, the following approach was adopted: A literature review on antibiotic degradation was carried out using multiple databases, namely Google Scholar, ScienceDirect, Web of Science, and PubMed. Search terms included ATBs degradation mechanism, persulfate, sonocatalysis, Fenton oxidation, electrochemical oxidation, photolysis, AOPs, and antibiotics' applied both individually and in combination. The terms 'ATBs' and 'AOPs' were consistently employed as the primary

search criteria. In addition, Scopus was used to survey existing studies, with searches performed using the keywords “antibiotics degradation” and “advanced oxidation processes,” as illustrated in Fig. 1b. The Mendeley reference manager was used to compile all of the retrieved articles. Studies released up until January 2025 are included in this review.

1.2 Source and Transportation of Antibiotics in Wastewater

The supplementary file Text S1 provides a thorough discussion of the origin and movement of antibiotics in wastewater.

1.3 UN Sustainable Development Goal 2030: Antibiotics and Their Environmental and Health Implications

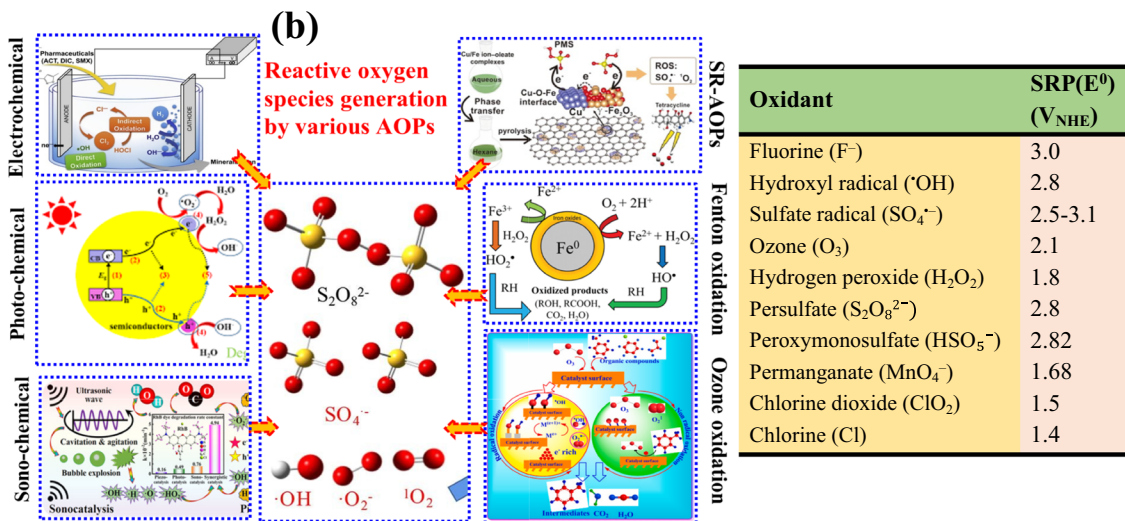
Antibiotics are widely used and stay in the environment for extended periods of time. They can also accumulate in organisms and as a result provide a significant barrier to protecting the long term viability and health of all aquatic ecosystems in the world (Aslam et al., 2024; Domingo-Echaburu et al., 2021). Antibiotic degradation is an important issue for meeting the United Nations Sustainable Development Goals (SDGs) 2030, especially Goal 3 (Good Health and Well-being) and Goal 6 (Clean Water and Sanitation) (Bhattacharya et al., 2024; Souza et al., 2021). Antimicrobial Resistance (AMR) a major threat to public health globally is driven by improper disposal of the drugs and environmental degradation. Human shedding, agricultural runoff, or disposal of pharmaceutical waste can introduce antibiotics into the environment and lead to the development of resistant bacteria. This contamination of water supplies and decreased efficacy of life-saving medications affects both ecosystems and human beings (Domingo-Echaburu et al., 2021).

Tackling this problem necessitates the preparation of plans for the proper disposal of antibiotics; improvement of wastewater treatment practices; and promotion of new technologies, such as AOP, particularly with SR-AOP, as part of the global efforts to achieve the UN Sustainable Development Goals (SDGs) by 2030, which encapsulate the need to protect ecosystems; maintain resilient health care systems; and ensure clean and safe access to sufficient

fresh water for all people. Pharmaceuticals may cause significant and complex negative consequences for both the environment and human health. The impact of pharmaceutical pollution on water quality and human health is presented in Fig. 2a. Scientific studies have demonstrated that even at low concentrations, pharmaceuticals can cause considerable harm to aquatic environments, including endocrine disruption, reproductive abnormalities, and antibiotic-resistant bacteria; as well as toxic effects on many of the various representatives of aquatic organisms (Domingo-Echaburu et al., 2021). Figure 2a illustrates that a variety of different classes of pharmaceutical pollutants are evidenced to cause significant harm to water bodies and associated ecosystems. These pollutants can enter the environment through multiple routes, including agricultural runoff, improper disposal, and excreta from human and animals. These pollutants can persist in soil and water; thereby potentially negatively affecting wildlife and ecosystems. For example, they may disrupt the endocrine system of aquatic organisms (Sosa-Hernández et al., 2021). The presence of pharmaceutical residues in drinking water supplies is harmful to human health, since they may disrupt endocrine systems and promote the development of drug-resistant microorganisms. To minimize the dangers of ATB discharge, it is necessary to implement effective waste management practices, utilize proper disposal methods, and employ new technologies for the treatment of water.

1.4 Antibiotic Remediation Policy Measures, Environmental Guidelines, and Regulatory Frameworks

To combat the rising problem of antimicrobial resistance (AMR), it is critical that there are international policies and regulations established to reduce antibiotics in wastewater. The purpose of these regulations is to limit the discharge of antibiotics from multiple sources including homes, hospitals, farms, and pharmaceutical companies into their respective waters. Various international agencies, national governments, and regulatory bodies have developed regulations in place to help manage and mitigate antibiotic contamination in wastewater. The international frameworks, as exemplified by the United Nations Environment Program (UNEP) and the World Health Organization (WHO), also stress the importance of using an

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integrated approach that contains a public education campaign coupled with the use of technology and regulatory standards.

1.4.1 International Guidelines and Policies

The WHO global action plan on antimicrobial resistance includes strategies to reduce antibiotic pollution in the environment (Khouja et al., 2022). It also includes water quality and wastewater management standards and suggests that measures should be put in place to reduce antibiotic pollution. The United Nations Environment Programme (UNEP) has developed frameworks for managing pharmaceutical pollutants, including antibiotics, in the environment with the assistance of other global organizations (Souza et al., 2021). The UNEP identifies the need for improved legislation concerning the release of pharmaceuticals and better wastewater treatment processes (Puri et al., 2023). The Organization for Economic Co-operation and Development (OECD) has also published reports and guidelines regarding the control of contaminants of growing concern, such as antibiotics, in wastewater (Miettinen & Khan, 2022). These documents outline legislative measures such as monitoring, effective treatment technologies, and pollution control. European Union (EU) regulations: The Water Framework Directive (WFD) is designed to preserve the qualitative and quantitative integrity of all bodies of water. Antibiotics are one of the many pollutants regulated under the WFD to ensure their monitoring and reduction. The Urban Wastewater Treatment Directive provides standards that dictate the collection, treatment, and discharge of municipal wastewater (Khouja et al., 2022). These standards relate to reducing environmental pollutants such as pharmaceuticals. The USEPA oversees discharge of wastewater per the Clean Water Act's regulations. USEPA has guidelines for proper pharmaceutical disposal, and they provide recommendations for reducing antibiotic water pollution. U.S. Food and Drug Administration (FDA) provides guidelines for safe disposal of pharmaceuticals to protect water quality from contamination (Puri et al., 2023). The National Pollutant Discharge Elimination System (NPDES) permit program manages the release of pollutant materials into U.S. bodies of water due to point source releases, which may include discharges of antibiotic compounds. Other countries have adopted a

variety of national action plans to address Antimicrobial Resistance (AMR), including controls on antibiotics in wastewater. The Chinese Ministry of Ecology and Environment has developed regulations for the discharge of pharmaceutical wastewater. The Government of India has developed guidelines for managing pharmaceutical waste to prevent creating environmental contamination. The Central Pollution Control Board (CPCB) has developed effluent discharge regulations that prohibit the presence of antibiotic residues in effluents (Kotwani et al., 2023). Environment and Climate Change Canada (ECCC) develop and administer laws, regulations and guidelines to prevent the environment from becoming contaminated with pharmaceutical products. The Government of Canada supports R&D for advanced wastewater treatment processes to eliminate the presence of antibiotics. In June 2015, the Norwegian government introduced a new national strategy (2015–2020) that set a clear goal of reducing antibiotic use in human medicine by 30% by 2020 compared with 2012 levels (Puri et al., 2023).

2 Mechanistic Insights into Antibiotic Degradation Strategies via SR-AOPs

SR-AOPs have emerged as a highly effective approach for the degradation of ATBs in water treatment. These processes depend of the formation of sulfate radicals ($\text{SO}_4^{\bullet-}$), which are powerful oxidizing agents capable of breaking down a wide range of organic pollutants, including persistent antibiotics (Arvind Kumar et al., 2024a, 2024b; Arvind Kumar et al., 2023a, 2023b). These processes utilized the oxidant materials such as peroxymonosulfate (PMS, HSO_5^-) or peroxydisulfate (PDS, $\text{S}_2\text{O}_8^{2-}$) also known as persulfate (PS) ions, leads to generating the $\text{SO}_4^{\bullet-}$ radicals (Iqbal et al., 2024). The main source of PDS is sodium persulfate ($\text{Na}_{2\text{DFTGF}}\text{S}_2\text{O}_8$), potassium persulfate ($\text{K}_2\text{S}_2\text{O}_8$), and ammonium persulfate ($(\text{NH}_4)_2\text{S}_2\text{O}_8$), whilst the common source for PMS is potassium monopersulfate triple salt (KHSO_5 , 0.5 KHSO_4 , 0.5 K_2SO_4), which is traded as oxone. As shown in Fig. 2b, these two persulfates are structurally similar to H_2O_2 , and both contain O–O bonds. The standard reduction–oxidation potential of $\text{SO}_4^{\bullet-}$ is high (Arvind Kumar et al., 2021a, 2021b), close to or even exceeding that of $\text{OH}^{\bullet}$ ($\text{SO}_4^{\bullet-}$: $E^0 = 2.5\text{--}3.1$ V, $\text{OH}^{\bullet}$: $E^0 = 1.9\text{--}2.7$ V), which

can theoretically degrade most organic pollutants (Dos Santos et al., 2023; Kang et al., 2019). Many investigations have shown the notable effectiveness of SR-AOPs, accomplishing almost complete degradation of ATBs in aqueous media (Arvind Kumar et al., 2024a, 2024b; Arvind Kumar et al., 2023a, 2023b; P. Liang et al., 2021). Several SR-AOP techniques have been established for wastewater contaminant removal, including UV irradiation, photocatalysis, ozonation, Fenton-based reactions, as well as sonochemical and electrochemical oxidation methods, as illustrated in Fig. 2b. These approaches initiate the generation of highly reactive oxygen species, including superoxide radical ($O_2^{\bullet-}$), singlet oxygen (1O_2), sulfate radical ($SO_4^{\bullet-}$), hydroxyl radical ($^{\bullet}OH$), and hydroperoxyl radical ($HO_2^{\bullet}$), capable of attacking on the ATBs molecule and dis-integrating the ATBs molecules into smaller, less harmful compounds, and ultimately, mineralizing into water, carbon dioxide, and inorganic ions (Ghanbari & Moradi, 2017; M. Shen et al., 2022; Zhu et al., 2020). The application of SR-AOPs

in ATBs degradation is gaining attention due to their efficiency, the possibility of tailoring the process to target specific contaminants, and their potential to alleviate the environmental impact of antibiotic pollution in water bodies (Arvind Kumar et al., 2023a, 2023b; Arvind Kumar et al., 2023a, 2023b).

Depending on the nature and concentration of ATBs in wastewater, various combinations of these methods may be used. Reactive oxygen species (ROSs) can be generated or accelerated by energy inputs like electrical power, ultrasonic irradiation, and ultraviolet light. These radicals are used by procedures including ozonation, photocatalysis, and Fenton-based systems to break down the intricate chemical structures of ATBs. Therefore, the persistence of ATBs in aquatic settings can be significantly reduced by using AOPs, lowering their potential ecological concerns and negative health impacts (Dos Santos et al., 2023). As shown in Eq. (1), AOPs typically consist of two primary stages: first, they produce powerful oxidants, and then they interact with ATB molecules in water.

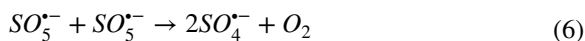
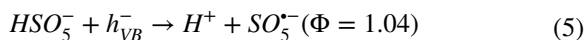
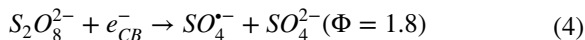
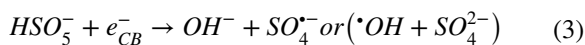
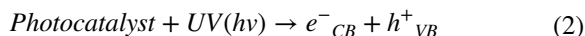


An integrated framework for eliminating ATBs from wastewater is supported by each AOP strategy's unique modes and performance levels (Ricardo et al., 2021). The next sections go into detail about several methods of ATB degradation.

2.1 SR-Driven Photolysis and Photocatalytic Degradation of ATBs

SR-mediated photocatalytic degradation has emerged as an effective strategy for eliminating ATBs through the use of semiconductor photocatalysts activated under suitable light irradiation, such as ultraviolet (UV) or visible light (Bu et al., 2024; Zheng et al., 2023). Due to its high irradiation energy, UV light irradiation is very effective at activating PDS ($S_2O_8^{2-}$) and PMS (HSO_4^-). By generating more reactive radicals for the breakdown of ATBs, the addition of PMS/PDS to the photocatalysis system improves the separation of electron-hole pairs and increases overall efficiency (Ma et al., 2023). A semiconductor produces photo-induced electrons and holes when it absorbs UV or visible light (Eq. 2). PMS/PDS can then trap

the electrons, limiting charge carrier recombination and generating $SO_4^{\bullet-}$ and $^{\bullet}OH$ radicals (Eqs. 3–4) (Diao et al., 2022; J. Wang et al., 2022). Sulfur pentoxide radicals ($SO_4^{\bullet-}$) (Ma et al., 2023) (Eq. 5), can be formed by the created holes reacting with PMS. These radicals may then react to make more $SO_4^{\bullet-}$ (Eq. 6).



Due to their high molar extinction coefficient (Φ), oxidants $S_2O_8^{2-}$ and HSO_5^- have O–O bonds that efficiently absorb UV light (Arvind Kumar et al., 2023a, 2023b; Zou et al., 2023). Which leads to homolytic

cleavage caused by this absorption, extremely reactive radicals such hydroxyl radicals ($\cdot\text{OH}$) and sulfate radicals ($\text{SO}_4^{\cdot-}$) are created, with the latter having a larger quantum yield (Xiong et al., 2021). The efficiency of this cleavage depends on factors like the O–O bond length, bond energy, and molar extinction coefficient of the peroxides (Isari et al., 2021). Among these, $\text{S}_2\text{O}_8^{2-}$ has a lower O–O bond length and bond energy, combined with a high molar extinction coefficient, leading to a higher yield of reactive radicals and, consequently, greater degradation efficiency. The yield of reactive radicals aggressively attacks on ATBs, leading to rupture the molecular structure of ATBs, and produce the intermediate compound and subsequently mineralization into carbon dioxide (CO_2) and water (H_2O) (Xiong et al.,

2021; Zheng et al., 2023). Therefore, the integration of PMS and PDS with photocatalysts is thus a highly effective strategy for enhancing the degradation of various ATBs, making it a promising approach for environmental remediation. Experimental conditions and removal efficiencies of ATBs by photocatalytic oxidation techniques tabulated in detailed in supplementary file Table S1.

In a recent investigation, (Diao et al., 2022), a team explored the mechanism of PMS-assisted photocatalytic degradation of norfloxacin (NOF) using a calcium-based Ag_3PO_4 composite photocatalyst under visible light irradiation as shown in Fig. 3a. There were two main reactions involving in NOF degradation by $\text{Ca-Ag}_3\text{PO}_4/\text{PMS}/\text{visible light}$ process, including radicals and non-radical oxidation reactions. In

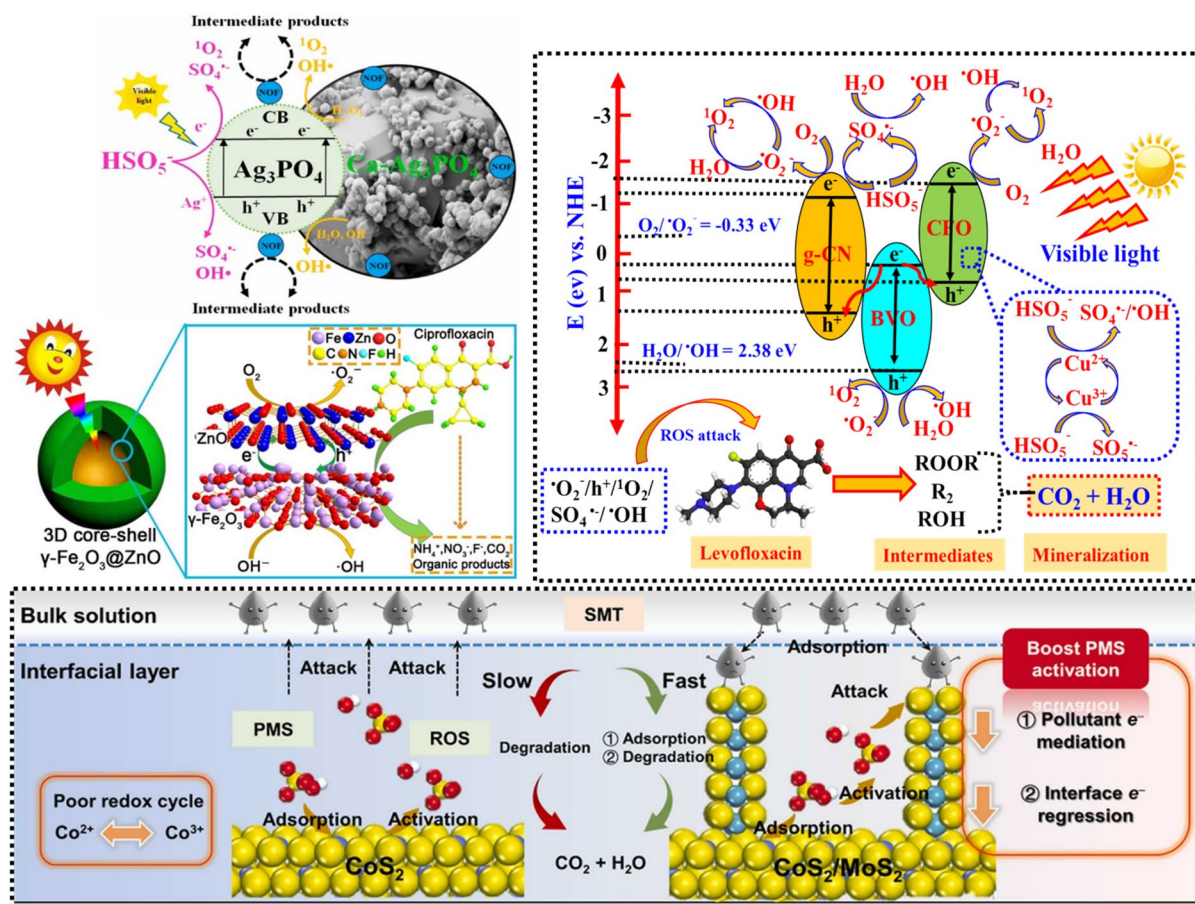
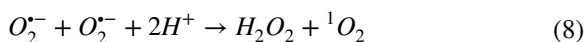
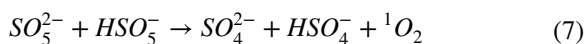


Fig. 3 Photocatalytic degradation mechanism of ATBs; **a** Norfloxacin degradation by $\text{Ca-Ag}_3\text{PO}_4/\text{PMS}$ under visible light (Diao et al., 2022; Li et al., 2017), **b** Levofloxacin degradation using Z-Scheme $\text{BiVO}_4\text{-Ag}_2\text{O}@g\text{-C}_3\text{N}_4$ (Li et al.,

2023a, 2023b, 2023c), **c** CIP degradation using 3D core-shell $\gamma\text{-Fe}_2\text{O}_3@\text{ZnO}$ photocatalysts and **d** SMT degradation by $\text{CoS}_2/\text{MoS}_2$ (Fang et al., 2024)

radical reaction, the potential ROS generated in the Ca-Ag₃PO₄/PMS/visible light process were HO•, SO₄•⁻ and ¹O₂, all took part in NOF degradation by, but both HO• and SO₄•⁻ was the primary species for NOF degradation. Additionally, Ag ions also released from Ca-Ag₃PO₄ during the reaction, these formed Ag ions further effectively activate PMS to produce SO₄•⁻, and some of the formed SO₄•⁻ could be converted into HO•. The generated radical (HO•, and SO₄•⁻) attacked on NOF molecule and cleave the structure and produce the intermediate and further mineralized to CO₂ and H₂O. In the non-radical oxidation reaction, ¹O₂ could be generated from the self-decomposition of PMS and reaction of H₂O₂ with the superoxide intermediates via Eqs. (7–8).



Photocatalytic activation of PMS can enhance the oxidation potential of the system by producing multiple reactive species and improving the separation of photogenerated charges through PMS-mediated electron trapping.

Another study by authors, combining photocatalytic processes with PMS activation represents a viable approach for eliminating levofloxacin (LEV) (Xufei Li et al., 2023a, 2023b, 2023c). Dual Z-scheme g-C₃N₄/BiVO₄/CuFe₂O₄ (CN/BVO/CFO) heterojunction photocatalyst activated with PMS under visible light system effectively removal of 96.2% of LEV within 1 h, as shown in Fig. 3b. Authors also proposed the photocatalytic mechanism, given the band alignment between CN, BVO and CFO components, two potential charge migration pathways (type-II and Z-scheme) may exist within the CN/BVO/CFO composite. Each catalyst constituent including BVO, CN and CFO can be stimulated to produce electron-hole pairs through visible light exposure. Meanwhile, adding PMS additionally enhanced the catalytic performance including LEV degradation efficiency, reaction kinetics and mineralization degree. PMS activation can occur through Cu⁺ or Fe²⁺ species (active sites in CN/BVO/CFO heterojunction) to produce ROS (h⁺, •O²⁻, •OH, SO₄•⁻ and ¹O₂). Ultimately, through the synergistic effects of ROS, LEV underwent decomposition into smaller compounds, inorganic salts, CO₂ and H₂O within the CN/BVO/CFO/Vis/PMS

reaction system. Researchers Feng et al., enhanced PMS activation utilizing composite photocatalyst SrTiO₃/BaFe₁₂O₁₉ for tetracycline removal under visible irradiation. Tetracycline removal efficiency achieved 96.1% in 60 s with reaction rate constant of 0.241 s⁻¹. This removal efficiency and reaction rate were 1.39 and 1.83 fold greater than the SrTiO₃/PMS/Vis reaction system, respectively. The Fe and Ti centers in the composite were determined as primary activation sites for PMS and H₂O activation, respectively. The h⁺, e⁻, and •O²⁻ exhibited dominant roles in TC removal (Feng et al., 2023). Another study by (Mandal et al., 2022) investigated the persulfate activated novel MoS₂/Bi₂MoO₆ composite photocatalyst, which was further sensitized using graphene quantum dots (GQDs), for ATB ofloxacin degradation under visible-light irradiation. The excellent photocatalytic performance of ofloxacin was noticed along with Cr(VI) reduction in aqueous solution under the combined effects of photocatalyst activated by PMS under visible-light. Under the visible light, pristine Bi₂MoO₆ and MoS₂ in 5 GQD-30 MoS₂/BMO were excited to the photogenerated holes in the VB and electrons (e⁻) in the CB. One of the major advantage of this system is that it has large surface area, which promotes the formation of reactive catalytic centers for ofloxacin adsorption, subsequently enhancing the catalytic mechanism by enabling accelerated surface interactions compared to those found in single material networks. GQDs within the composite catalyst structure serve various functions; a primary function involves broadening the visible-light absorption capacity of 5 GQD-30 MoS₂/BMO, while another involves facilitating electron migration within the 30 MoS₂/BMO catalytic framework which enhanced the degradation of ofloxacin.

One of the interesting study by authors Li et al., visible-light-driven PMS activation by novel bimetallic Co-Mo-TiO₂ nanomaterial for sulfamethoxazole (SMX) degradation (J. Li et al., 2023a, 2023b, 2023c). Approximately 100% SMX removal occurred within 30 min using Vis/Co-Mo-TiO₂/PMS system, with its reaction kinetic constant (0.099 min⁻¹) being 24.8 fold greater than the Vis/TiO₂/PMS system (0.014 min⁻¹). The oxidation-reduction processes of Co³⁺/Co²⁺ and Mo⁶⁺/Mo⁴⁺ within the enhanced Co-Mo-TiO₂/PMS produced diverse ROSs throughout PMS activation while both ROS ¹O₂ and SO₄•⁻ served as primary reactive species for SMX

removal. Furthermore, density functional theory (DFT) calculations indicated that Co–Mo–TiO₂ demonstrated strong affinity toward PMS adsorption, evidenced by the O–O bond distance from PMS and the E_{ads} values of the catalysts. A recent breakthrough in photocatalyst design for solar-driven sulfate radical-based advanced oxidation processes (SR-AOPs) was achieved through the fabrication of a magnetically separable by 3D core–shell γ -Fe₂O₃@ZnO as shown in Fig. 3c (N. Li et al., 2017). The ZnO shell thickness was controlled such that it resulted in a high surface area and high surface potential, which led to higher photocatalytic activity under solar illumination for the composite material compared to the reference γ -Fe₂O₃ photocatalyst. The composite material degraded ciprofloxacin at a rate of 92.5%, while the γ -Fe₂O₃ degraded ciprofloxacin at a rate of 11.7% under the same simulated solar irradiation condition. The presence of a type II heterojunction resulted in increased separation of photo-generated electron–hole pairs that resulted in a greater production of hydroxyl radicals, hydroxyl radicals and holes as electron acceptors resulting in kinetic rate constants of 4–20 times greater than their respective pristine materials. Therefore this research illustrates the potential of heterostructure creation via interfacial engineering to create photocatalysts with enhanced antibiotic degradation efficiencies and contributes to the logical development of multi-functional photocatalysts for use in water remediation technologies.

Recent advances in catalytic materials and constrained nanoreactors have improved the efficacy and robustness of surface redox-activated oxygen photodissociation (SR-AOP). One of the more substantial uses of interface reengineering to fabricate CoS₂/MoS₂ heterostructures, shown in Fig. 3d (Fang et al., 2024), is to simultaneously enhance both the amount of reactive oxygen species produced during a heterogeneous Fenton-like reaction, such as SO₄^{•−} radicals, and also maintain a stable electrical configuration at the interface of the two materials through strong Co–S–Mo interactions.

In the case of confined nanoreactors, the CoS₂/MoS₂ system confines antibiotics such as sulfamethazine while increasing the electron transfer rate due to a reduction in distance between the reactive species produced (namely ROS) and the reaction site. The result of these two advantages is a very high degradation rate (0.302 min^{−1}) along with outstanding

long-term operational stability (93% activity after 8 cycles) across a wide pH range (3–11). Additionally, the special synergy between MoS₂ and CoS₂ enables multi-electron transport pathways, where interfacial electron regression increases PMS activation and MoS₂ encourages valence cycling of Co ions. In the end, these coordinated mechanisms result in effective and lasting SMT degradation by speeding up SO₄^{•−} production.

2.1.1 Bond Cleavage Mechanistic Insight of ATBs by Photocatalytic Action and Degradation Pathways

The rapid development of computational chemistry makes it possible to predict the reactive sites of a relatively large molecule of ATBs (Arvind Kumar et al., 2024a, 2024b; D. Zhang et al., 2020). Destruction of the chemical structures and bond breakage information of typical ATBs during photocatalytic action caused by the attacking the ROSs on the ATBs molecules are well summarized, and the results are listed in Fig. 4. Authors (J. Li et al., 2023a, 2023b, 2023c) identified the potential reaction sites of SMX molecules vulnerable to nucleophilic attack (f^+), free radical attack (f^0) and electrophilic attack (f^-) using Fukui functions by DFT calculations. Fukui function is a popular tool for predicting the region selectivity in a radical involved reaction. Based on the Fukui functions results, produced ROS attacked on the 4 potential reactive sites and destruct the molecule structure as shown in Fig. 4, (1) S–N bond cleavage by ROS attacked, (sulfonamide S–N bond is highly susceptible to oxidative cleavage by [•]OH and h⁺, disrupting the core sulfonamide moiety). (2) oxazole ring cleavage (attack by ROS leads to ring-opening reactions in the isoxazole ring, resulting in fragmentation and increased molecular reactivity), (3) [•]OH radical attack on aniline ring and cleavage the aniline ring (The aromatic aniline moiety undergoes hydroxyl radical-mediated oxidation, causing ring cleavage and formation of phenolic intermediates), and (4) detachment of amino group by oxidation of aniline moiety (Oxidative deamination of the aniline group occurs, contributing to loss of functional integrity) (J. Li et al., 2023a, 2023b, 2023c; Yangju Li et al., 2020; X. Shen et al., 2024; Zhou et al., 2024). All the deformation of SMX molecule produced the intermediate compound which were further oxidized by attacking the ROSs and mineralized into CO₂ and H₂O.

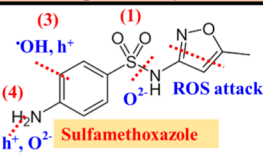
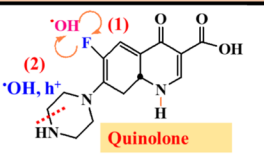
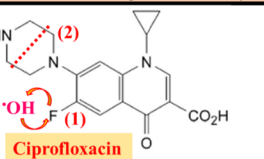
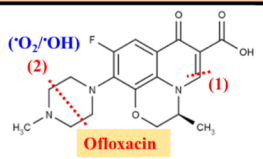
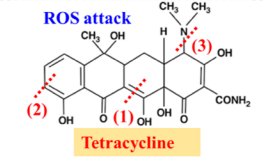
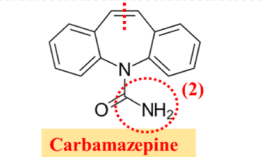
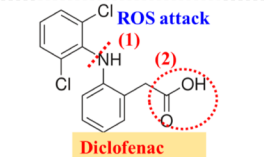
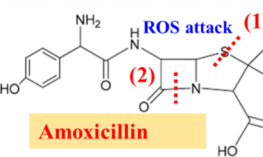
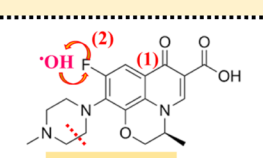
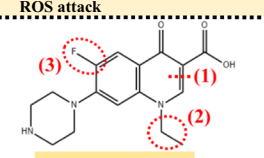
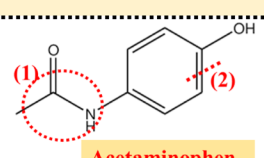
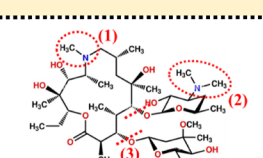
SR-based photocatalysis	SR-based electrochemical	SR-based heterogeneous catalysis	SR-based sonochemical
 Sulfamethoxazole (1) S-N bond breakage (2) Oxazole ring cleavage (3) $\cdot\text{OH}$ attack on aniline ring (4) Oxidation of aniline moiety	 Quinolone (1) F replace by $\cdot\text{OH}$ radical (2) Piperazine ring cleavage	 Ciprofloxacin (1) F replace by $\cdot\text{OH}$ radical (defluorination) (2) Piperazine ring cleavage and dialdehyde form	 Ofloxacin (1) Nucleophilic radicals ($\cdot\text{O}_2/\cdot\text{OH}$) attack on tetrahydropyridine ring (2) Cleavage of N-piperazine ring by ROS and leaving amino group
 Tetracycline (1) C-S bond breakage (2) Hydration process by ROS attack (3) Deamination by ROS attack	 Carbamazepine (1) Central heterocyclic ring cleavage attacked by ($\cdot\text{O}_2/\cdot\text{OH}$) due to high frontier electron density (2) Detachment of amide group by ROS attack	 Diclofenac (1) C-N bond breakage (2) -COOH group detached by ROS attack	 Amoxicillin (1) ROS attack on S thioether group (2) Quaternary ring cleavage by ROS (3) ROS attack (decarboxylation)
 Levofloxacin (1) $\cdot\text{OH}$ attacking on piperazine ring (2) F replace by $\cdot\text{OH}$ radical (defluorination)	 Norfloxacin (1) $\cdot\text{OH}$ attacking on C=C bond in quinolone ring (2) ROS attack on C=C bond adjacent to carboxyl group form after ring cleavage (3) Breakage of C-F bond by $\cdot\text{OH}$ radical (defluorination)	 Acetaminophen (1) Cleavage of N-H bond by $\cdot\text{OH}$ attacking (Amino-oxidation) (2) Ring opening by $\cdot\text{OH}$ attack on C-C bond (Hydroxylation)	 Azithromycin (1) $\cdot\text{O}_2/\cdot\text{OH}/\text{h}^+$ attacked on C-N bond (2) $\cdot\text{O}_2/\cdot\text{OH}/\text{h}^+$ attacked on -N- bond (demethylation) (3) Separation of D-desosamine or L-cladinose ring by ROS attack

Fig. 4 Molecule structural destruction and bond breakage of typical ATBs (i.e., sulfamethoxazole, quinolones, ciprofloxacin, tetracycline, carbamazepine, diclofenac, amoxicillin, levofloxacin, norfloxacin, acetaminophen, and azithromycin)

The another investigation by authors (J. Wang et al., 2022) explored the tetracycline degradation under visible-light-driven photocatalysis of g-C3N4/FeWO₄ Z-scheme heterojunction activated by PMS. Tetracycline undergoes several bond breakage processes under photocatalytic conditions, which are started by ROSs (i.e., $^1\text{O}_2$, h^+ , $\text{O}_2^{\cdot-}$, $\cdot\text{OH}$) attacking the TC molecule. (1) The conjugated ring system's C=C double bonds are susceptible to ring opening due to oxidative cleavage by $\cdot\text{OH}$ radicals. (2) ROS attack increases molecule polarity and instability by promoting hydration processes, particularly on carbonyl and alkene groups. (3) ROS-induced oxidative breakage of C-NH₂ bonds causes deamination, which removes amino groups and disrupts functional

by attacking the ROSs produced from various SR-AOPs (e.g., photocatalytic, electrochemical, heterogeneous catalysis, and sono-chemical oxidation)

moieties (Feng et al., 2023; Shi et al., 2022; J. Wang et al., 2022). Molecular fragmentation and the production of intermediate compounds are the results of these successive reactions. These compounds are then further oxidized and eventually mineralized into CO₂, H₂O, and inorganic ions. Levofloxacin is degraded under photocatalytic conditions mainly through two important routes. The electron-rich piperazine ring is first attacked by $\cdot\text{OH}$ radicals, which cause it to cleave and compromise the structural integrity of the medication.

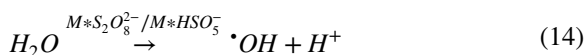
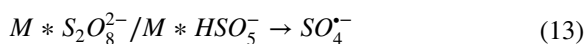
Second, a defluorination process takes place in which $\cdot\text{OH}$ nucleophilically substitutes a hydroxyl group for the fluorine atom on the quinolone core (Xufei Li et al., 2023a, 2023b, 2023c).

2.2 SR-Based Electrochemical Oxidation of ATBs

SR-based electrochemical oxidation (SR-EO) represents a highly promising approach for eliminating ATBs from aquatic environments (Kiendrebeogo et al., 2022). Electrochemical oxidation processes are typically conducted in an electrochemical cell with two electrodes: anode and cathode (Yuan et al., 2023). When electric energy is applied to the electrochemical cell, oxidation reaction occurs at the anode and generating ROS ($\bullet\text{OH}$, $\text{O}_2^{\bullet-}$, $\text{HO}_2^{\bullet}$, $^1\text{O}_2$), while a reduction reaction takes place at the cathode, producing hydrogen. Subsequently, ROS attacked to EC molecules and oxidize them. When external oxidants like PMS or PDS are introduced into the electrochemical cell, they produce additional ROS, such as $\text{SO}_4^{\bullet-}$, which enhances the electrochemical oxidation of ECs. SR-EO tend to accumulate the oxidant (PMS, PDS and PS) at the anode surface through electrostatic forces, where it experiences direct or indirect electrochemical activation processes. The experimental conditions and removal efficiencies of ATBs by electrochemical oxidation techniques tabulated in detailed in supplementary file Table S2. The particular activation mechanism and reactive intermediates formed during electrochemical treatment depend on the chosen electrode materials. During direct electro-oxidation, adsorbed PDS or PMS species at the electrode surface can experience direct transformation into transition state forms specifically $\text{M}^*\text{S}_2\text{O}_8^{2-}$ or M^*HSO_5^- (where M denotes anode surface) (Eqs. (9) and (10)) [64]. Unlike the stable $\text{M}^*\text{S}_2\text{O}_8^{2-}$ and M^*HSO_5^- species, the transition configurations $\text{M}^*\text{S}_2\text{O}_8^{2-}$ and M^*HSO_5^- exhibit a highly reactive condition. This non-radical form represents a strongly active electrophile. When ATBs molecules approach the electrode interface, they function as electron donors whereas the transition state forms function as electron acceptors, enabling electron transfer processes. Through this mechanism, the ATBs molecules surrender electrons and undergo degradation whereas the transition state converts into sulfate species Eqs. (11–12) (R denotes the ATBs undergoing degradation in the system. The "intermediate" chemical intermediates represent an unspecified form of oxidized ATBs "R" molecules) (H. Song et al., 2018).



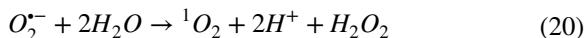
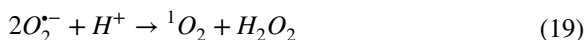
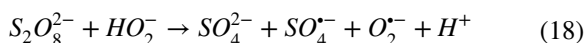
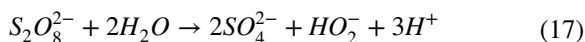
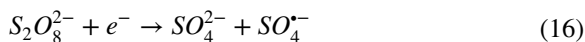
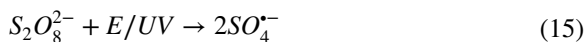
Additionally, a portion of $\text{M}^*\text{S}_2\text{O}_8^{2-}$ and M^*HSO_5^- can additionally experience transformation into $\text{SO}_4^{\bullet-}$ (Eq. (13)) (Xiaobao Li et al., 2022). Based on density functional theory (DFT) computations, the activation energy for PDS transformation to produce $\text{SO}_4^{\bullet-}$ without an electrical field is 0.63 eV, while the complete energy barrier for PDS formation of a transition state form and subsequent conversion to $\text{SO}_4^{\bullet-}$ is merely 0.16 eV. The DFT findings indicate that PDS can be more easily activated at the electrode surface to produce $\text{SO}_4^{\bullet-}$ under an electrical field (Shao et al., 2022).



Furthermore, the transition forms $\text{M}^*\text{S}_2\text{O}_8^{2-}$ and M^*HSO_5^- promote water electrolysis, allowing the production of $\bullet\text{OH}$ (Eq. 14) (Fu et al., 2022). Song et al. contrasted the electrochemical activation of PDS and PMS with individual electro-oxidation procedures, and the resulting dissolved oxygen (DO), chronoamperometry (CA), and cyclic voltammetry (CV) analyses revealed that $\text{M}^*\text{S}_2\text{O}_8^{2-}$ and M^*HSO_5^- formed at the electrode interface could promote water electrolysis to yield additional $\bullet\text{OH}$ radicals (H. Song et al., 2018). Earlier research has concluded from these electrochemical characterization results that PDS produces highly active transition state intermediates ($\text{M}^*\text{S}_2\text{O}_8^{2-}$) at the anode interface (C. Song et al., 2017; H. Song et al., 2018). Unfortunately, there exists presently a shortage of efficient techniques for direct and quantitative detection of the transition state configurations $\text{M}^*\text{S}_2\text{O}_8^{2-}$ and M^*HSO_5^- (Shao et al., 2022; H. Song et al., 2018).

During indirect oxidation, PMS or PDS interacts with reactive intermediates (e.g. $\bullet\text{OH}$) produced at the anode surface and is indirectly transformed to $\text{SO}_4^{\bullet-}$. Indirect electrochemical activation involves

the interaction of adsorbed species M ($\bullet\text{OH}$) formed on the electrode interface with PMS/PDS to produce $\text{SO}_4^{\bullet-}$, or the interaction of metallic ions produced by sacrificial anodes with PMS/PDS to produce $\text{SO}_4^{\bullet-}$ (Xiaobao Li et al., 2022; Ren et al., 2024). Both indirect and direct electrochemical activation of PS occur simultaneously at the electrode surface. And the characteristics of the electrode material dictate whether direct or indirect electrochemical activation of PS prevails at the electrode interface. Another recent study by (B. Wang & Wang, 2024), degradation of carbamazepine by electrochemical-pressure UV-activated peroxodisulphate process. System generate the ROSs such as $^1\text{O}_2$, $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$, but $^1\text{O}_2$ was the dominant species for the degradation of carbamazepine. Degradation of carbamazepine by $^1\text{O}_2$. $^1\text{O}_2$ can be produced in two ways. One is the production of $\text{O}_2^{\bullet-}$ by hydrolysis of PDS, as shown in reaction in Eqs. (17) and (18), the produced $\text{O}_2^{\bullet-}$ reacts with H^+ to create $^1\text{O}_2$, as presented in reaction in Eq. (19). Additional breakdown products of PDS hydrolysis, such as HO_2 and HO_2^- , are oxidized by $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$ to create $^1\text{O}_2$, as presented in Eq. (20).



The time evolution of anode materials for electrochemical degradation of pharmaceuticals in wastewater reflects significant advancements as shown in Fig. 5a. From noble-metal electrodes like Pt and Ru (1983–2012), research shifted to B-doped diamond (2003–2013), then to mixed metal oxides (MMO) such as Ti/Pt/SnO₂–Sb₂O₄ and Ti/Ti₄O₇ (2013–2018) (Nabgan et al., 2022). In 2022, TiO₂ nanotube arrays (TNA) anode and Cu–Co–WO₃ cathode emerged. By 2024, cerium-doped Sb₂S₃ anodes were introduced, offering enhanced performance. In 2025, cutting-edge

materials such as Ti₄O₇, Ti/TiO₂/βPbO₂ electrodes, and nickel silicide nanowires are under exploration, promising superior efficiency and stability in the degradation of persistent pharmaceutical contaminants.

2.2.1 Bond Cleavage Mechanistic Insight of ATBs by Electrochemical Action and Degradation Pathways

A wide range of ATBs have complex cyclic chemical frameworks, including beta-lactam, bicyclic, and tetracyclic ring systems. Examples of these include quinolones, carbamazepine, tetracyclines, and amide alcohols. When these ring structures are broken down by electrochemical advanced oxidation processes (EAOPs), a number of transformation intermediates are produced (Fig. 4). A number of reactions occurs prior to ring breakage, including as the removal of oxygen-bearing functional groups and the substitution or removal of substituents. During the electrochemical oxidation of quinolones, $\bullet\text{OH}$ radical assault and the ensuing structural changes are the main mechanisms of bond breakdown.

A hydroxylated intermediate is first formed when a highly reactive $\bullet\text{OH}$ radical replaces the fluorine atom on the quinolone ring. The molecule is more vulnerable to further oxidative destruction as a result of this substitution. The cleavage of the piperazine ring, a structural characteristic shared by many quinolone ATBs, is one of the crucial subsequent steps. The attack of $\bullet\text{OH}$ radicals or other reactive oxygen species causes this ring-opening process, which fragments the molecular structure and reduces antibiotic effectiveness (Jia et al., 2024).

A novel study led by Li et al. found that carbamazepine (CBZ) can be effectively destroyed using electrochemical oxidation with nitrogen-rich porous carbon material decorated with an FeCo alloy (FeCo@NC). This study demonstrates that the use of electrically-enhanced PMS (E-FeCo@NC/CF-PMS) to activate CBZ degradation is possible and effective. As shown in Fig. 5b, in-situ generated ROS's specifically attack areas of high frontier electron density (FED), with the $\bullet\text{OH}$ and superoxide ($\text{O}_2^{\bullet-}$) radicals being the main drivers of bond breakage as illustrated by Fig. 4, (Li et al., 2023a, 2023b, 2023c). The olefin double bond in the central heterocyclic ring of CBZ has a very high FED and therefore is very susceptible to ROS attack resulting in the formation of

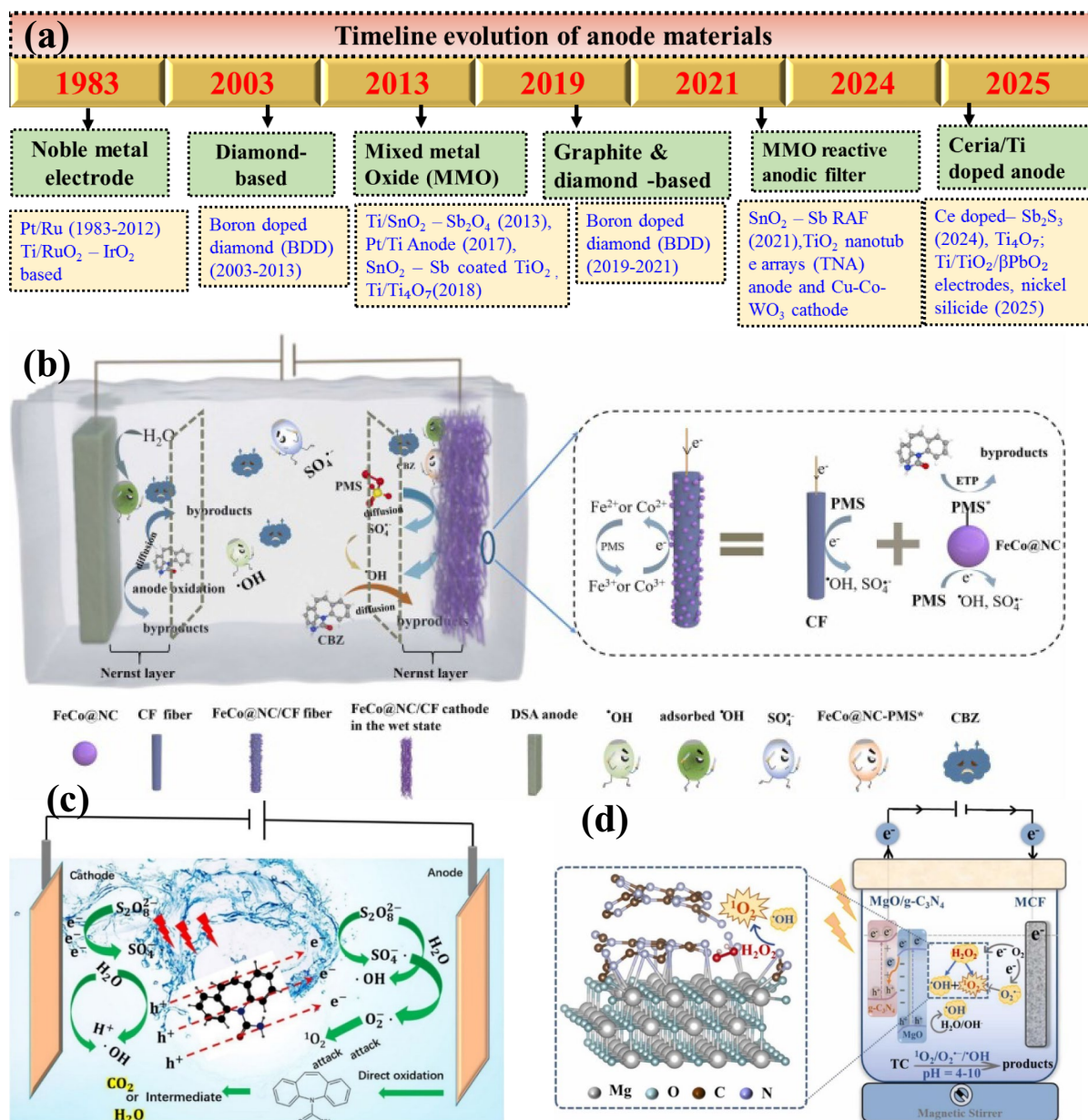


Fig. 5 **a** Chronological development of materials employed in fabricating anode electrodes for electrochemical degradation of pharmaceuticals in wastewater (Nabgan et al., 2022); **b** Proposed mechanism for CBZ degradation via electro-activation of FeCo@NC/CF-PMS system (Li et al., 2023a,

2023b, 2023c); **c** Degradation mechanism pathways of carbamazepine via E/UV/PDS system (Wang & Wang, 2024); **d** Schematic representation of photo-electrocatalytic mechanism of tetracycline using S-scheme MgO/g-C₃N₄ photoanode (Sun et al., 2024)

10,11-epoxycarbamazepine as an initial intermediate. After that the core structure (central heterocyclic ring) will break leading to the formation of a dialdehyde molecule. Furthermore, ROS cause the amide group to separate from the heterocyclic core, which

advances the CBZ breakdown process. Reactive oxygen species (ROS), especially hydroxyl radicals ($\cdot\text{OH}$), enhance a number of bond breakdown processes involved in the electrochemical oxidation of norfloxacin (Carneiro et al., 2020; Wang et al., 2021).

(1) The first stage entails $\bullet\text{OH}$ radicals attacking the quinolone ring's $\text{C}=\text{C}$ link cause the ring to open and the molecular structure to become unstable. (2) After ring cleavage, ROS attack the $\text{C}=\text{C}$ bond next to the carboxyl group, further breaking down the molecule and upsetting its functional groups. (3) Lastly, the strong $\text{C}-\text{F}$ bond is cleaved by $\bullet\text{OH}$ radicals, which causes defluorination and the creation of less lasting breakdown products. During electrochemical treatment, these successive processes play a major role in the breakdown and mineralization of norfloxacin.

As seen in Fig. 5c (Wang & Wang, 2024), recent developments in SR-AOPs have demonstrated the exceptional effectiveness of electrochemical oxidation-assisted UV-activated peroxodisulfate (E/UV/PDS) systems in antibiotic degradation. For the degradation of carbamazepine, a comparison of PDS, E/PDS, UV/PDS, and E/UV/PDS systems showed a substantial synergistic relationship, reaching up to 99% elimination in 120 min. The mechanistic study indicated that $^1\text{O}_2$ was the primary reactive species in the SR-AOP system, followed by $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$ radicals. The process's efficacy was governed by parameters such as the PDS dose, applied voltage, initial pollutant concentration and pH. As applied voltage increased, the production of $\text{SO}_4^{\bullet-}$ and H_2O_2 also increased; thus, the degradation potential was enhanced through an increase in the kinetics of degradation as well. In acidic conditions, $\text{SO}_4^{\bullet-}$ was the primary radical species produced through electrochemical reduction, while in alkaline conditions $\bullet\text{OH}$ was the dominant radical species produced; in neutral conditions, there was an equal contribution from both $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$ radicals. Therefore, there is molecular-level evidence of the chemistry of the interfacial reaction in addition to radical-driven pathways of the SR-AOP process; thus, the use of E/UV/PDS to treat persistent pharmaceutical contaminants in aqueous matrices is a viable remediation strategy. The photo-electrocatalytic (PEC) system described above demonstrated superior performance in the treatment of tetracycline compared to the use of an amine-scheme $\text{MgO/g-C}_3\text{N}_4$ photoanode and a modified carbon felt (MCF) cathode for tetracycline removal. After just 30 min of visible light irradiation under -0.5 V vs. Ag/AgCl , the tetracycline was completely removed from the aqueous matrix (see Fig. 5d) (Sun et al., 2024).

2.3 Sulfate Radicals Based Heterogeneous Catalysis Process for ATBs

Sulfate radicals based heterogeneous catalytic systems (SR-HCs) have developed as a viable approach due to their superior performance and capacity to mineralize ATBs into non-toxic products (Fan et al., 2023; Kumar et al., 2024a, 2024b). This process utilized the oxidizing agents like PMS, PDS and PS that produce the highly reactive ROS such as $\text{SO}_4^{\bullet-}$, $\bullet\text{OH}$, $\text{O}_2^{\bullet-}$, $\text{HO}_2^{\bullet}$, and $^1\text{O}_2$ as shown in Fig. 6a (Kumar et al., 2024a, 2024b; Liang et al., 2023). Produced ROS attacking on the ATBs molecule and dis-integrate the ATB molecules into smaller, less harmful compounds, and ultimately mineralization into water, carbon dioxide, and inorganic ions, as detailed mechanism depicted in Fig. 6a. The direct interaction of PMS and PDS with ATBs is constrained by their slow reaction kinetics (Hassani et al., 2023). Hence, PMS and PDS need an appropriate activation method to produce potent reactive oxidizing intermediates such as $\text{SO}_4^{\bullet-}$, and $\bullet\text{OH}$. The activation efficacy of PMS and PDS depends on the catalytic materials and the active sites on their surface (Liang et al., 2023). Typically, $\text{SO}_4^{\bullet-}$, can be generated by stimulating different PMS and PDS compounds through O–O cleavage using electron transfer or external energy. The energy-driven activation of PMS and PDS can be achieved through thermal treatment, UV irradiation, ultrasonic (US) methods, electrochemical techniques (Kumar et al., 2023a, 2023b). In the previous year's various types of catalytic materials have been employed for activation of PMS/PDS for the degradation of ATBs. Single transition metals-based catalysts, mixed metals-based catalysts (layered double hydroxides (LDHs), layered double oxides (LDO), perovskites), carbon material (granular activated carbon, graphene functionalized, biochar functionalized), metal organic framework (MOF), covalent organic framework (COF), Zero-valent metal activation, etc., (Kumar et al., 2021a, 2021b; Kumar & Prasad, 2021; Lu et al., 2018). Among the activation techniques, the utilization of transition metals represents a conventional and efficient approach for PMS/PDS activation. The mechanism of PMS/PDS activation involves electron transfer from transition metals to persulfate demonstrated in Fig. 6a. The reduction process takes place in the interaction of persulfate

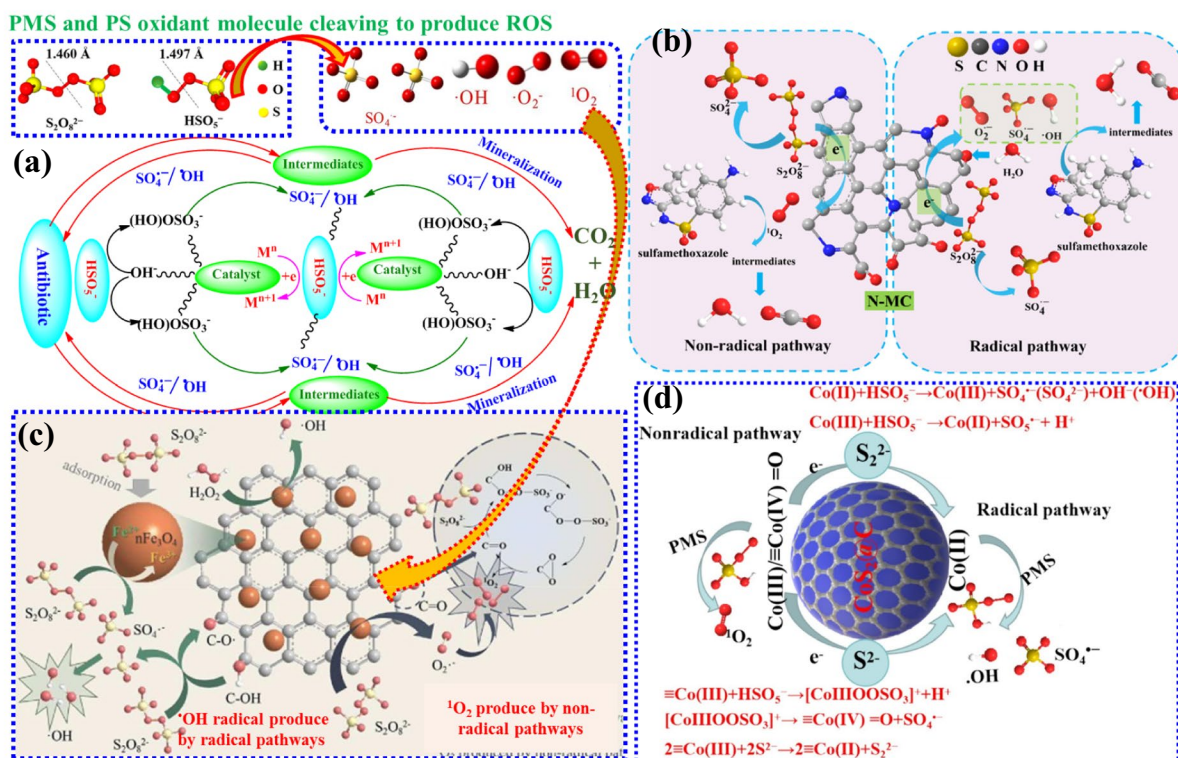
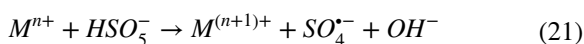


Fig. 6 **a** Proposed mechanism of ROSs generation during PMS and PDS activation for ATBs degradation (Kumar et al., 2023a, 2023b), **b** SMX degradation by radical and non-radical pathways via PDS activation (Zhang et al., 2024a, 2024b), **c**

Proposed mechanism of Fe-LBH to active PDS for ATBs degradation (M. Wang et al., 2023a, 2023b), **d** Possible activation mechanism of PMS by radical and non-radical pathways using $CoS_2@C-3$ (Li et al., 2025)

with transition metals according to the following Eqs. (21–22).



As Anipsitakis and Dionysios confirmed, Ag(I) represents the optimal activator for PDS while Co(II) species exhibited superior performance for PMS activation. These findings have been linked to the redox potential of transition metals where Co(II)/Co(III) possesses elevated redox potential (1.92 V) which exceeds that of PMS (1.82 V) (Ghanbari & Moradi, 2017). Nevertheless, the redox potential is not the sole factor for the activation process. For example, there exists a distinction between PMS and PDS during activation utilizing transition metals, where PMS can additionally function as an electron donor when exposed to a higher oxidation state (Mn^{+1}) of

transition metals (Fe^{3+} , Co^{3+} , Cu^{2+}) and generate PMS radicals or sulfur pentoxide radical ($SO_5^{\bullet-}$), via the cleavage of H–O bond (Kumar et al., 2023a, 2023b; Peng et al., 2020). While PDS cannot be degraded by Mn^{+1} , this might be attributed to the symmetrical molecular configuration of PDS (Liang et al., 2023). Therefore, numerous researchers argued that PMS has preference over PDS for the interaction with transition metals (Hassani et al., 2023; Kumar et al., 2024a, 2024b). In our recent study Kumar et al., perovskites $LaMO_3$ ($M=Fe, Co, Cu$) showed the remarkable PMS activation efficacy (97.11%) for the degradation of ofloxacin ATBs (Arvind Kumar et al., 2023a, 2023b).

Perovskite $LaCoO_3$ /PMS system generate the various ROSs ($SO_4^{\bullet-}$, $O_2^{\bullet-}$, $\cdot O_2$, $\cdot OH$), but ROS species ($\cdot OH$ and $SO_4^{\bullet-}$) were dominant species for ofloxacin degradation. Recent studies have revealed that antibiotic degradation via SR-AOPs involves both radical and non-radical oxidative pathways,

offering enhanced mechanistic understanding. In the case of sulfamethoxazole (SMZ), nitrogen-doped mesoporous carbon (NMC) has emerged as a highly effective persulfate (PS) activator, as displayed in Fig. 6b (Zhang et al., 2024a, 2024b). The activation triggers cleavage of the O–O bond in $S_2O_8^{2-}$, producing $SO_4^{\bullet-}$ radicals through interactions with electron-rich nitrogen species (N-py, N-pyr, N-g) and sp^2 carbon. These radicals further generate $HO^{\bullet}$ from water. Simultaneously, in the non-radical pathway, PS reacts with H_2O forming $O_2^{\bullet-}$, which combines with $HO^{\bullet}$ to yield singlet oxygen (1O_2). The deep oxidative mineralization of SMZ into CO_2 , H_2O , and other intermediates is facilitated by the cooperation of $SO_4^{\bullet-}$, $HO^{\bullet}$, and 1O_2 . The significance of interfacial chemistry and the characteristics of active sites on NMC in improving SR-AOPs performance is highlighted by this dual-pathway degradation. The logical development of catalysts for the effective elimination of persistent antibiotics is guided by these mechanistic discoveries. The synergistic mechanisms behind PDS activation in SR-AOPs for antibiotic degradation have been clarified by recent research, especially in Fe-LBH-PDS systems, as shown in Fig. 6c (Wang et al., 2023a, 2023b). PDS is efficiently activated by iron-loaded biochar (Fe-LBH) through radical ($^{\bullet}OH/SO_4^{\bullet-}$) and non-radical (1O_2) routes. Oxygen-containing functional groups ($-C-OH$, $-C=O$) and Fe(II) species in Fe_3O_4 operate as active sites, encouraging PDS adsorption and the subsequent breakage of the O–O bond. Tetracycline (TC) degradation greatly outperformed pristine biochar systems, surpassing 85% in 30 min. Concurrent breakdown of TC, inactivation of antibiotic-resistant bacteria (60.4%), and decrease of ARGs up to 2.74 $\log_2$ -fold are made possible by the dual processes. The dominance of $^{\bullet}OH$ and 1O_2 was confirmed by EPR and scavenging tests. These results demonstrate how Fe-LBH-PDS systems can use specific radical and non-radical interfacial interactions to reduce antibiotics and resistance elements in wastewater. In addition, the details, experimental conditions and removal efficiencies of ATBs by heterogeneous catalytic oxidation with various catalytic material have been tabulated in supplementary file Table S4.

As seen in Fig. 6d, recent developments in SR-AOPs have brought attention to the catalytic efficiency of metal sulfide–carbon composites made from MOF precursors. $CoS_2@C$ composites made

with ZIF-67, Co-MOF-71 and Co-MOF-74 are a prominent example; $CoS_2@C$ -3 (from Co-MOF-74) exhibits superior PMS activation ($E_a = 25.62$ kJ/mol) (Li et al., 2025). Within ten minutes, all antibiotics (CIP, MNZ, LVF, and NFT) were completely degraded. Both radical ($SO_4^{\bullet-}$, $^{\bullet}OH$) and nonradical (1O_2 produced via $\equiv Co(IV)=O$, electron transfer) routes drove this process. Through improved electron transport and in situ $\equiv Co(II)$ regeneration, the sulfur moiety promoted PMS activation. Excellent pH tolerance, HA resistance, and recyclability were shown by the system with more than 70% efficiency over 5 cycles.

2.3.1 Bond Cleavage Mechanistic Insight of ATBs by Heterogeneous Catalysis and Degradation Pathways

The destruction of antibiotic molecular structures and bond breakage through heterogeneous catalysis using PMS/PDS activation is a promising approach to eliminate antibiotics from aqueous media (Honarmandrad et al., 2023). The heterogeneous catalyst mechanism typically involves transition metals or metal oxides to produce highly reactive ROSs, which attack the ATB molecules, leading to the abstraction of hydrogen atoms, addition to unsaturated bonds, and cleavage of C–C, C–N, and other crucial bonds within the antibiotic molecules (Arvind Kumar et al., 2024a, 2024b; Liang et al., 2023). In the recent study by (Ji et al., 2024) identified the potential reaction sites of amoxicillin molecules vulnerable to nucleophilic attack (f^+), free radical attack (f^0) and electrophilic attack (f^-) using Fukui functions by DFT calculations. Higher attack index values indicated a greater likelihood of CIP molecules being attacked by the corresponding types of species. Ciprofloxacin bond breakage mechanism involves multiple oxidative steps. (1) The initial attack by highly reactive ROSs i.e., $SO_4^{\bullet-}/^{\bullet}OH$, generated from PMS/PDS, leads to the substitution of the fluorine atom by a $^{\bullet}OH$, resulting in defluorination of the ciprofloxacin molecule. (2) Subsequently, the piperazine ring undergoes oxidative cleavage, forming intermediate structures that further degrade into dialdehyde compounds (Huang et al., 2020; Ji et al., 2024; Li et al., 2025).

The sequence of transformations illustrates that $SO_4^{\bullet-}$ radicals are used to degrade an ATB's complex structure by removing covalently-bonded portions of

the molecule stepwise. Recently, (Jiang et al., 2024) described the mechanism of acetaminophen bond breakage, which occurs via two primary steps: (1) a sulfate radical breaks the N–H bonding via $\bullet\text{OH}$ radicals attack leading to amino-oxidation where the nitrogen–hydrogen is cleaved to create a means to form a stable intermediate. (2) Secondly, the aromatic ring's conjugated C=C link is attacked by hydroxyl radicals ($\bullet\text{OH}$), which causes ring opening through hydroxylation (Qutob et al., 2022). The aromatic structure is broken by this reaction, producing compounds with different chemical characteristics. Under sulfate radical catalysis, both mechanisms aid in the conversion and breakdown of acetaminophen. Similarly, Fig. 4 depicts the diclofenac bond process. The cyanide (CN) link in the diclofenac molecule is first broken by radical attack, which is started by sulfate radicals produced during the catalytic step. The CN bond degrades as a result of these radicals' reaction with it. Secondly, ROS-induced oxidative stress causes the carboxyl ($-\text{COOH}$) group of diclofenac to

separate (Gao et al., 2022). By attacking the $-\text{COOH}$ group, these ROS encourage its extraction from the parent structure. Under oxidative conditions, both pathways aid in the transformation and mineralization of diclofenac.

2.4 SR Based Sono-Chemical Oxidation of ATBs

SR based sono-chemical oxidation is a promising approach for degrading ATBs, employing ultrasonic irradiation to generate intense localized environments that break down structure of ATBs molecule (Bretas Alvim et al., 2021). Sono-chemical advanced oxidation processes utilize the ultrasound irradiation (typically 20–1000 kHz) which induce the acoustic cavitation (AC) in water. AC is the phenomenon of micro bubbles formation, growth, and implosive collapse in the liquid as depicted in Fig. 7a. The rapid collapse of micro bubbles produces high-frequency ultrasound waves, create hot spots with extreme local temperatures (~ 5000 K) and pressures (~ 1000 atm), which leads to

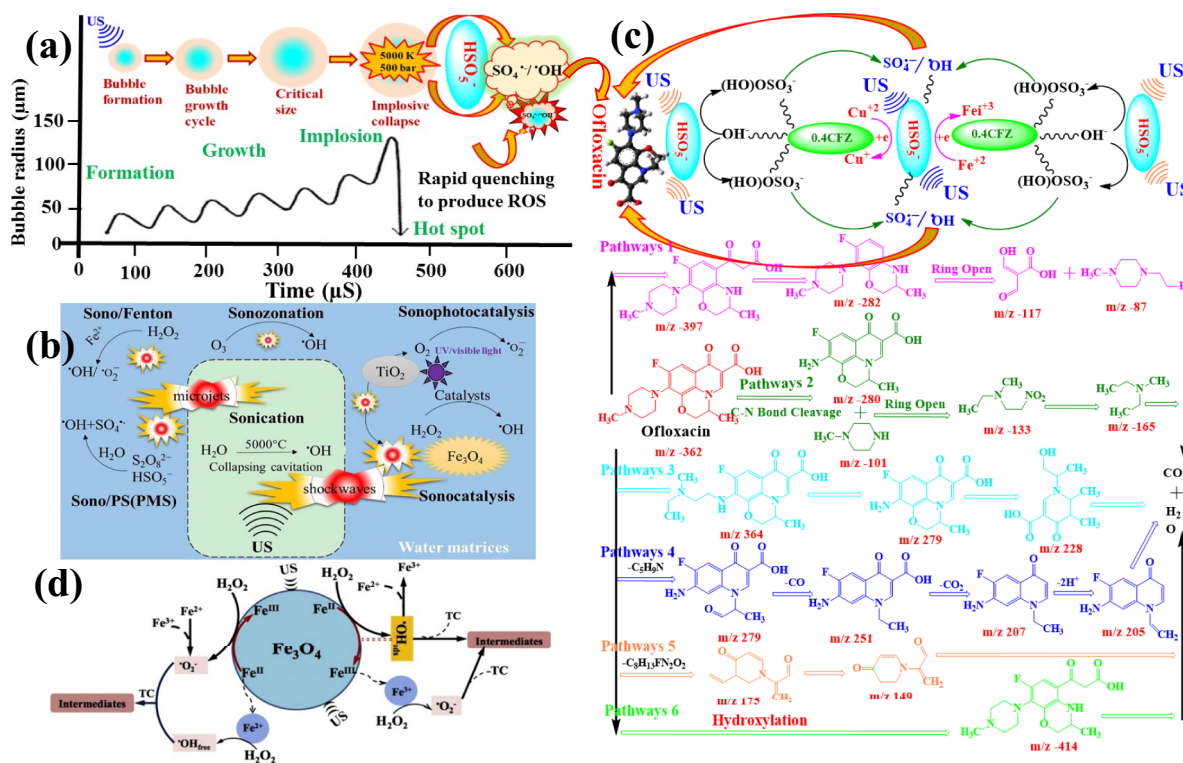


Fig. 7 a Ultrasound cavitation bubble formation mechanism simultaneously ROS generation, b, d ultrasound induced sonocatalytic degradation of tetracycline mechanism (Hou et al.,

2016; Liu et al., 2021a, 2021b), and c ultrasound induced PMS activation mechanism using series $\text{Cu}_x\text{Fe}_{1-x}\text{ZnO}$ LDO for ofloxacin degradation (Kumar et al., 2023a, 2023b)

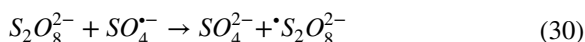
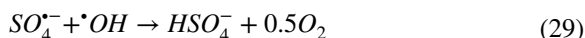
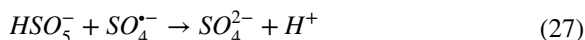
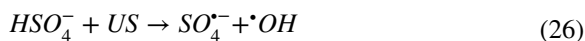
the formation of ROSs i.e., $\bullet\text{OH}$, $^1\text{O}_2$, superoxide radical ($\text{O}_2^{\bullet-}$), and hydroperoxyl radical ($\text{HO}_2^\bullet$) (Cai et al., 2020; Kumar et al., 2023a, 2023b). While introducing external oxidants like PMS or PDS, it produces the additional ROS like $\text{SO}_4^{\bullet-}$ radicals, which enhances the sono-chemical oxidation of ECs. The process of radical generation and following degradation includes multiple stages. Initially, water molecules inside the imploding cavitation bubbles decompose due to the extreme conditions, creating hydroxyl and hydrogen radicals via Eq. (23):



These radicals subsequently migrate into the adjacent solution and engage with the ATBs. The extremely reactive $\bullet\text{OH}$ radicals target the carbon-carbon bonds within the ATBs, simpler molecules. This can be expressed through the reaction in Eq. (24):

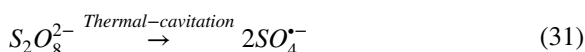


Here, RH denotes a portion of the ATBs molecule, and $\text{R}^\bullet$ represents a hydrocarbon radical that can additionally interact with oxygen to produce different oxidation compounds. Sonication not only provides a novel approach for the elimination of persistent organic contaminants, but also enhances SR generation from the interactions of PS and PMS with $\bullet\text{OH}$ that is produced in situ during sonication, as demonstrated in Eqs. (25–30). Thus, the SR and $\bullet\text{OH}$ that are generated from the activation of PS or PMS under sonication have been regarded as the sources of antibiotic degradation.



Next, the breaking of chemical bonds within antibiotic molecules, including S–N, S–C and N–C of

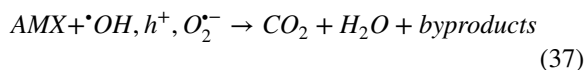
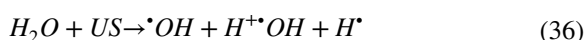
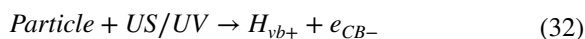
SMZ (Zhang et al., 2021), or the elimination of the N-methyl, hydroxyl and amino groups of TC takes place through oxidation with SR and $\bullet\text{OH}$ (Nimshi et al., 2023). Elevated temperature significantly enhanced the cavitation activity and chemical effects, leading to greater degradation rates of ATBs under Sono/Oxone or sono/ $\text{S}_2\text{O}_8^{2-}$ processes (Liu et al., 2021a, 2021b). PS activation can be achieved through thermal energy to generate SR, as demonstrated in Eq. (31) (Kumar et al., 2024a, 2024b).



The activation of PMS and PDS in Sono/PS(PMS) systems, hydrogen peroxide in Sono-Fenton, and ozone in sonozonation combined with the physical effects of cavitation, such as microjets and shockwaves significantly enhance mass transfer and catalyst activation (Hou et al., 2016; Liu et al., 2021a, 2021b). In sonocatalysis and sonophotocatalysis, the combination of catalysts (e.g., Fe_3O_4 , TiO_2) and UV/visible light further intensifies reactive oxygen species (ROS) production, improving ATB degradation efficiency in complex aqueous matrices as shown in Fig. 7b, d. Therefore, SR-AOPs are a potent platform for reducing persistent antibiotics in environmental fluids because this integrated approach improves radical production, surface reactivity, and molecular fragmentation.

In the recent by Hapeshi, et al. examined the degradation of OFC utilizing a 20 kHz horn-type ultrasonic reactor. The findings demonstrated that the RE of OFC rises with increasing US power density under the optimal conditions, 27.7 μM OFC and 640 W/L of US power density, the RE of OFC in a 350 mL solution achieved 31% after 240 min sonication (Hapeshi et al., 2013). Although the removal efficacy for OFC removal was limited only employing the US irradiation. Therefore, one more strategy we explored in our recent investigations (Arvind Kumar et al., 2024a, 2024b). We explored the OFC degradation by using the hybrid system $\text{Cu}_x\text{Fe}_{1-x}\text{ZnO}$ layer double oxide (LDO)/PMS/US designed for the degradation of OFC. There exist limited case studies specifically examining the degradation of ATBs via cavitation methods, and only a restricted amount of literature has investigated this field. For example, (Yang et al., 2024) recently utilized a sonocatalytic method employing ultrasonic irradiation to process

microplastics and dye, utilizing Ti-based MOFs supported on Ti_3C_2 MXene. A recent study by Al-Musawi et al., explored the sonophotocatalytic degradation mechanism of amoxicillin using $\text{Fe}_3\text{O}_4@\text{SiO}_2/\text{PAEDTC}$ surrounded by MIL-101(Fe) in aquatic environment (Al-Musawi et al., 2024). AMX molecules can be adsorbed onto the sonophotocatalyst interface. Simultaneously, the photocatalytic process begins upon exposing the semiconductor to UV radiation and generating electrons and h^+ within the conduction and valence bands (Eq. 32). The photoexcited electron can form $\text{O}_2^{\bullet-}$ by interacting with the O_2 molecule at the catalyst interface (Eq. 33), as verified by the scavenger findings. In contrast, h^+ interacts with H_2O and OH^- molecules to form the $\bullet\text{OH}$ (Eq. 34–35). The existence of ultrasonic waves within the photocatalytic reactor not only cleaves water molecules through collapse bubbles to generate reactive intermediates (Eq. 36) but also enhances the production of $\bullet\text{OH}$ by purifying the interface of the photocatalyst and improving active centers. Eventually, produced reactive intermediates, e.g., $\text{O}_2^{\bullet-}$, $\bullet\text{OH}$ and h^+ can break down AMX molecules into simple biodegradable compounds, CO_2 , and H_2O (Eq. 37).



2.4.1 Sono-Chemical Assisted Bond Cleavage Mechanism of ATBs, and Degradation Pathways

In our recent study, we explored the bond cleavage mechanism for OFC molecule by sono-chemical oxidation utilizing the PMS oxidant (Kumar et al., 2024a, 2024b; Liu et al., 2021a, 2021b). We identified all the reactive attacking sites by Fucui index

using DFT calculations (e.g., nucleophilic, electrophilic and free radical attacking sites on the OFC molecule). In addition, we proposed 4 potential degradation pathways for OFC degradation. In pathway I, the degradation of OFC initiated with hydroxylation on the tetrahydropyridine ring (Hapeshi et al., 2013; Su et al., 2022). Consequently, the cleaved ring was attacked by nucleophilic radicals ($\bullet\text{O}_2/\bullet\text{OH}$) at the C1 atom, which possessed the highest reactivity value (f^+ , 0.037). The area surrounding the C1 atom, shown in blue on the Hirshfeld surface Fig. 7c, displays the highest positive charge density, indicating that the C1 atom is especially vulnerable to nucleophilic radical attacks. In pathways II, C-N bond connecting the N-piperazine ring and benzene structure was broken when attacked by ROS species $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$, leading to the formation of intermediate products. Within the third pathway, the fragmented N-piperazine ring may potentially interact with radical species to generate intermediates of m/z 364, which were then subsequently broken down into compound with m/z values of 279 and 228 through cleavage of C-N bond in benzene ring (Kumar et al., 2024a, 2024b). The bond breakage mechanism underlying the sulfate radical-based sono-chemical oxidation of amoxicillin (AMX) involves sequential radical-induced transformations, as supported by DFT calculations (Al-Musawi et al., 2024). There have been two main degradation processes suggested. (1) In-situ generated $\text{SO}_4^{\bullet-}/\bullet\text{OH}$ radicals in the first pathway attacks the sulfur atom in AMX's thioether group to start the oxidation process, resulting in the production of a sulfoxide intermediate (Karim & Shriwastav, 2021). This intermediate is then hydroxylated to produce monohydroxy AMX sulfoxide. (2) The quaternary β -lactam ring is attacked by ROS in the second pathway, which cleaves it and produces penicilloic acid. Following decarboxylation and further oxidation, this product eventually becomes a derivative of penicilloic acid. The intermediate molecules mineralize into carbon dioxide and water during prolonged exposure to ROS (Al-Musawi et al., 2024; Karim & Shriwastav, 2021). Azithromycin is sonochemically oxidized by sulfate radicals through a number of important reactive oxidative mechanisms. (1) The azithromycin structure's C-N bond is first targeted by reactive species such $\bullet\text{O}_2/\bullet\text{OH}/h^+$, which cleaves it and destabilizes the molecular structure (Kumar et al., 2021a, 2021b). (2) Demethylation then takes place by directly attacking

the –N– bond by radicals $\bullet\text{O}_2^-/\bullet\text{OH}/\text{h}^+$, which eliminates methyl groups and modifies the functional groups essential for azithromycin's antibacterial activity (Ling et al., 2023). (3) Lastly, the macrolide structure is further broken down and its biological activity is greatly reduced when ROSs facilitate the separation of sugar moieties, either the D-desosamine or the L-cladinose ring. Furthermore, a comprehensive summary of experimental conditions, operational parameters, and corresponding antibiotic removal efficiencies achieved through various sono-chemical oxidation processes, highlighting the influence of reaction conditions on degradation performance are listed in Table S3., in supplementary file.

Overall, the degradation pathways of ATBs during SR-AOPs are strongly dependent on the activation strategy, generated reactive species, and molecular structure of the target contaminants. In photocatalytic SR-AOPs, photo-induced electrons/holes and PMS/PDS activation generate ROS ($\text{SO}_4\bullet^-$, $\bullet\text{OH}$, $\text{O}_2\bullet^-$, and $^1\text{O}_2$), which primarily induce selective oxidation at electron-rich functional groups. For example, SMX undergoes S–N bond cleavage, isoxazole ring opening, hydroxyl radical-induced aniline ring degradation, and oxidative deamination, whereas TC degradation proceeds through C=C bond cleavage, ring-opening reactions, hydration, and deamination. Similarly, LEV and CIP degradation mainly involve $\bullet\text{OH}$ -mediated piperazine ring cleavage and defluorination. In electrochemical SR-AOPs, degradation is governed by direct electron-transfer reactions at electrode interfaces and electrochemically generated ROS, where bond cleavage is associated with oxidation at highly electron-rich regions. For instance, CIP and NOR undergo sequential defluorination, quinolone ring cleavage, and piperazine ring destruction through $\bullet\text{OH}$ attack, while CBZ degradation involves oxidation of the central heterocyclic C=C bond, formation of epoxy intermediates, ring-opening, and amide group detachment. In heterogeneous catalytic SR-AOPs, transition-metal catalysts, metal oxides, perovskites, LDHs, MOFs, and carbon-based materials activate PMS/PDS through interfacial electron transfer, producing radical and non-radical oxidants. For example, OFLX degradation over LaMO_3 /PMS systems involves hydroxylation, piperazine ring cleavage, and C–N bond breaking, while SMX degradation through N-doped carbon/PDS and tetracycline degradation using Fe-functionalized biochar/

PDS systems proceed through sulfonamide bond cleavage, aromatic ring oxidation, deamination, and multiple ROS-mediated fragmentation pathways. In sonochemical SR-AOPs, ultrasonic cavitation generates localized extreme conditions that promote PMS/PDS activation and enhance radical formation, leading to rapid molecular fragmentation. For example, OFLX degradation involves hydroxylation of the tetrahydropyridine ring, C–N bond cleavage between the piperazine and aromatic structures, and subsequent fragmentation of intermediates; AMX undergoes thioether oxidation, β -lactam ring cleavage, and decarboxylation, whereas AZM degradation occurs through C–N bond cleavage, demethylation, and detachment of sugar moieties. Overall, although $\text{SO}_4\bullet^-$ and $\bullet\text{OH}$ are common oxidizing species across all SR-AOP systems, the dominant degradation pathways are determined by the antibiotic molecular architecture, ROS selectivity, and activation mechanism, highlighting the importance of molecular-level understanding for designing efficient and sustainable ATB remediation strategies.

3 Toxicity Assessment of ATBs and their Intermediates Products by ECOSAR

The toxicity assessment of antibiotics (ATBs) and their degradation intermediates is essential for evaluating their potential environmental risks (Hassani et al., 2023; Liang et al., 2023). The Ecological Structure Activity Relationships (ECOSAR) model is a widely recognized predictive tool that estimates the toxicity of chemical compounds to aquatic organisms based on their molecular structures. In this review, the toxicity profiles of various ATBs and their corresponding degradation products are summarized in Fig. 8a–e. Numerous studies have reported the generation of diverse transformation products (TPs) during the degradation of different classes of emerging contaminants ECs (Wang et al., 2023a, 2023b). In our previous work, the aquatic toxicity of OFLX and its degradation intermediates was systematically evaluated using the ECOSAR model. The results predicted both acute and chronic toxicity toward representative aquatic organisms, including fish, daphnids, and algae, thereby providing valuable insights into the environmental safety of the degradation process (Kumar et al., 2024a, 2024b).

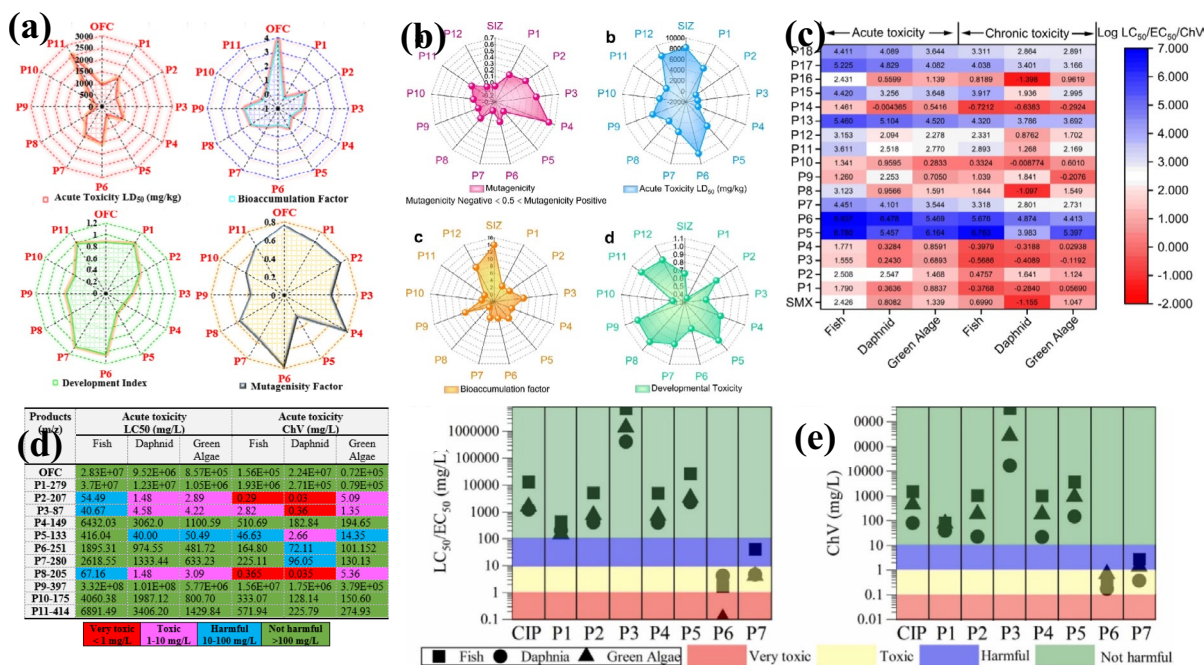


Fig. 8 Predicted the toxicity of ATBs and its intermediates by the T.E.S.T. for ECOSAR. **a, d** toxicity of ofloxacin (Kumar et al., 2024a, 2024b; Wang et al., 2023a, 2023b), **b, c, e** toxic-

ity of sulfamethoxazole and ciprofloxacin (Liang et al., 2023; Zhang et al., 2023)

The ECOSAR results revealed that OFC exhibits moderate toxicity towards all three organisms, with its chronic toxicity being of particular concern, as results shown in Fig. 8a, d. Compared to the original parent compound (OFC), some of the 11 identified degradation intermediates, including P2, P3, and P8, showed greater chronic toxicity to aquatic creatures like fish and daphnids. Compounds like P2 and P3 were shown to be more detrimental to daphnia and green algae in terms of acute toxicity. However most of the intermediates had lower toxicity to OFC indicating a decreasing level of ecological risk associated with their degradation process. The analysis supports the use of QSAR-based assessments as a means to support the finding that the degradation methods used in the 0.4CFZ-LDO/PMS/US system are capable of reducing the environmental hazard of OFC through detoxification and ultimately converting them to non-hazardous end products such as carbon dioxide and water.

In a recent work (Wang et al., 2023a, 2023b), evaluated the mutagenicity, acute toxicity, bioaccumulation potential, and developmental toxicity of ciprofloxacin and its breakdown intermediates using

ECOSAR. With very few exceptions, the majority of degradation intermediates showed decreased mutagenicity when compared to the parent molecule, as seen in Fig. 8e. In general, the intermediates' acute oral toxicity (rat LD₅₀) was higher than that of ciprofloxacin, indicating reduced acute toxicity; however, some intermediates, such as P3, showed noticeably lower LD₅₀ values, indicating enhanced toxicity. Furthermore, all intermediates had lower bioaccumulation values than ciprofloxacin, indicating a decreased capacity to accumulate in living things. Certain by-products showed enhanced developmental toxicity, whereas others showed decreased toxicity. Overall, the results indicate that although ciprofloxacin's catalytic degradation lowers overall toxicity, the creation of some hazardous intermediates is still an issue.

Acute toxicity tests and QSAR predictions were used to assess the toxicity of SMX and its breakdown intermediates (Zhang et al., 2023). When *Vibrio fischeri* inhibition was used to quantify acute toxicity, SMX's initial modest toxicity (~11%) increased dramatically after ozonation, particularly in the MB/O₃ process (up to 95.52%) as opposed to CB/O₃ (75.04%). This increase suggests that SMX itself

was not as hazardous as the intermediates created during breakdown. Through QSAR analysis, using ECOSAR, SMX is predicted to be an extremely toxic aquatic organism, especially the *Daphnia* that are located nearby. Hydroxylated intermediates such as P1, P3, P4, P9, and P10 appear to have more toxic characteristics than the SMX compound itself because they contain hydroxyl groups that changes the structural look of the benzene or isoxazole groups, as illustrated in Fig. 8c. The non-hydroxylated intermediates (e.g., P5, P6, and P7) are predicted to have decreased toxicity levels than that of the non-hydroxylated counterparts. When a ring structure becomes cleaved into smaller molecular sizes, there is a large reduction in toxicity. The experimental data along with the predicted results raised the awareness of the presence of potentially hazardous intermediates created during the ozonation of a compound and that toxicities from intermediate products should be evaluated in addition to any parent compound or parent compound degradation product being tested. Comprehensive overview of the ecotoxicological effects of different antibiotics on a variety of biological systems, emphasizing how they affect terrestrial species, aquatic creatures, microbes, and ecological health in the many environmental compartments described in Table S6 (supporting file).

4 Challenges and Future Research Directions for ATBs Mitigation by SR-AOPs

Aqueous environments can benefit greatly from the mitigation of aqueous contaminants (ATBs) through the use of SR-AOPs). The main challenges affecting the ability to apply and optimize SR-AOP technologies for practical use include high operational costs, complicated reaction mechanisms, and limited performance due to the diversity of water and wastewater matrices. The potential for transformation products to have toxicological effects as well as to generate secondary pollutants is still poorly understood, increasing the challenges for larger-scale implementation below are the challenges and proposed future research paths in advancing SR-AOPs development: Although there is strong promise in SR-AOPs for ATB mitigation, barriers to achieving their widespread use and efficacy exist. Wastewater matrices are highly complex and significantly impact the efficacy of

SR-AOPs, because inorganic ions and natural organic matter can act as scavengers for reactive oxygen species (ROS), while pH shifts can change the availability of ROS; **(a)** there is a high cost associated with the operation of SR-AOPs, which includes energy-intensive processes, specialized catalysts, and specialized oxidants that make the economically viable for extensive use; and **(b)** incomplete mineralization of ATBs frequently produces transformation products, some of which may have unknown or potentially greater toxicity than the parent ATBs, thereby creating a risk to both the environment and human health. **(c)** The challenge of scaling up from a laboratory testing environment to a real-world application for SR-AOPs is compounded by the issues of reactor design and consistency of reactions, which will be major barriers in the implementation of SR-AOPs at a large scale. **(d)** Optimizing the generation and control of reactive ROSs species (i.e., $\text{SO}_4^{\bullet-}$ and $\bullet\text{OH}$) is another significant challenge that must be addressed to improve overall degradation efficiency and reduce the potential for secondary pollution. These issues will need to be addressed in order to take SR-AOPs from an experimental approach to a viable and sustainable technology for remediating ATB.

Future research must focus on various areas that relate to addressing these problems: **(a)** Novel and cost-effective catalysts, with better reactivity and resistance to fouling, should be developed, as they will help improve the efficiency of degradation processes, thereby reducing operational costs; thus supporting the use of renewable energy sources (i.e., solar irradiation) as Sustainable alternatives to energy-consuming processes to increase the efficiency of systems; **(b)** Significant efforts must be made to conduct thorough studies on the degradation routes of ATBs to understand how they interact with other molecules and determine whether any toxic byproducts are produced in order to develop ways to reduce the formation of toxic byproducts; **(c)** New computational tools and bioassays will provide scientists with the means to fully assess the effects of the Intermediate Toxicity Associated with ATBs usage on the environment and human health; **(d)** to enhance real-world applicability, research should prioritize validating SR-AOPs in diverse wastewater matrices and optimizing process parameters, including oxidant dosage, reaction conditions, and energy inputs, to achieve consistent performance. In addition, novel

SR-AOP reactors that integrate conventional treatment technologies such as biological treatment will provide for solutions that are integrated and scalable. Reductions in costs through catalyst regeneration and reuse of oxidizers will continue to promote the economic viability of the AOPs. (e) Finally, researcher, industry and policy maker partnerships are critical to set forth criteria and guidelines to allow for the safe and effective implementation while mitigating the increasing environmental burden caused by the total quantity of ATB pollutants.

5 Concluding Remarks

ATBs have emerged as a critical environmental concern due to their extensive use in human and veterinary medicine, aquaculture, and agriculture. Their persistence in aquatic environments accelerates the proliferation of antibiotic-resistant bacteria (ARB) and genes (ARG), posing significant risks to ecosystem stability and public health. This review focused on the detailed insights into SR-AOPs, emphasizing their powerful capacity to break down persistent antibiotic pollutants via ROS production, transformation routes, intermediate product formation via molecular breakdown. SR-AOPs are an effective and durable way to reduce antibiotic contamination because of the exceptional oxidation strength, selectivity, and reactivity of $\text{SO}_4^{\bullet-}$ radicals. However, the main challenges are incomplete mineralization of ATBs, the potential for the production of toxic transformation products (i.e., AOP by-products), and the high cost of operating such systems, which limit their implementation on a large scale. Improving system performance will depend on optimizing operational parameters such as pH, oxidant dosage, and the design of the catalyst/designer to improve system performance. Future research should be focus on developing economical and sustainable catalytic materials are necessary to effectively integrate renewable energy and create innovative hybrid systems that enhance scalability and efficiency. Advanced analytical tools and computational modelling will assist in providing a greater understanding of the degradation processes and environmental consequences of by-products produced from SR-AOPs. In addition, the validation of SR-AOPs on a variety of wastewater matrices in real-world situations is an essential element toward

successful implementation. Collaboration among engineers, researchers, and policymakers in different disciplines will be critical to translating success in the laboratory into success in the real world. By addressing these needs, SR-AOPs can become a key technology that can help address the problems associated with ATB contamination, protect aqueous ecosystems, and promote global sustainability initiatives. SR-AOPs are a novel, sustainable means of degrading ATBs; however, significant advancements are required in order to overcome limitations currently impacting their pathway to economical use at industrial levels. If researchers and engineers work collaboratively across scientific and engineering disciplines in conjunction with policymakers, SR-AOPs may evolve into a key technology that ensures continued access to safe drinking water resources while supporting the advancement of global sustainability goals.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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